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Acquisition-aware multimodal multiple instance learning with cross-attention and hypergraph reasoning for examination-level multi-label classification of gastroscopy examinations: a retrospective multicenter study

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Abstract

Routine gastroscopy archives contain ordered images and finding descriptions, but conventional image-level learning analyzes selected images independently and loses examination-level context. We developed AMEF-MIL, an acquisition-aware multimodal multiple instance learning framework that integrates endoscopic images and gastroscopic finding descriptions for examination-level multi-label classification. The image pathway combines ConvNeXt-Tiny, APro-CoPE, label-wise attention MIL, and label hypergraph reasoning to encode up to 64 ordered images with acquisition-anchored, content-adaptive positions. The text pathway masks expressions that directly identify target categories to reduce label leakage and encodes finding descriptions with a multi-scale TextCNN. Label-query cross-attention retrieves disease-relevant text, gated residual fusion forms the primary multimodal output, and an auxiliary image-only branch predicts from visual representations. During training, multimodal branch supervision updates the shared visual encoder and a pretrained, frozen multimodal teacher provides knowledge distillation; the auxiliary branch uses only images at inference. Evaluation included 2861 examinations and 171591 images collected from Zhongshan Hospital, Fudan University, and Zhongshan Hospital (Xiamen), Fudan University, across white-light endoscopy, chromoendoscopy, surgical gastroscopy, and endoscopic ultrasonography, with targets for esophageal submucosal tumor, esophageal mucosal lesion, and gastritis and patient-grouped five-fold splits. Initial examination-level labels generated by rule-based matching of diagnostic conclusions were independently reviewed by two chief physicians, one from each participating hospital, each with 10 years of clinical experience in gastrointestinal endoscopy. The primary multimodal output achieved macro F1 scores of 0.9149, 0.8877, 0.8456, and 0.9064 on the four datasets, respectively, outperforming all compared image-only, text-only, and multimodal methods. Image-budget and position ablations supported APro-CoPE for ordered sequence modeling; ablation across all 16 combinations of the four main components supported the combined contribution of hypergraph reasoning, label-query cross-attention, and gated fusion; and multimodal branch supervision with knowledge distillation improved the auxiliary image-only output. These findings show that our proposed examination-level multimodal multi-label framework can jointly model multiple images acquired during one examination, paired finding descriptions, and coexisting diseases without image-by-image annotation, while retaining image-only prediction when finding-description text is unavailable.

Keywords

gastroscopy, examination-level multi-label classification, multiple instance learning, multimodal learning, acquisition-aware position encoding, knowledge distillation

中文摘要：常规胃镜临床归档数据包含有序图像和胃镜所见描述，但传统图像级学习独立分析筛选后的图像，因而丢失检查级上下文信息。本文提出 AMEF-MIL，一种融合内镜图像与胃镜所见描述、用于胃镜检查级多标签分类的采集感知多模态多实例学习框架。图像端结合 ConvNeXt-Tiny、APro-CoPE、多标签注意力 MIL 和标签超图推理，对最多 64 张有序图像进行采集锚定式内容自适应位置建模。文本端对能够直接指示目标类别的表述进行掩码以降低标签泄漏，并使用多尺度 TextCNN 编码胃镜所见描述。标签查询 cross-attention 检索疾病相关文本，门控残差融合形成主要多模态输出，辅助纯图像

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分支则根据视觉表征进行预测。训练阶段由多模态分支监督更新共享视觉编码器，并由预训练且冻结的多模态教师提供知识蒸馏；辅助分支在推理阶段仅使用图像。实验纳入来自复旦大学附属中山医院和复旦大学附属中山医院厦门医院的 2861 次检查和 171591 张图像，覆盖白光胃镜、染色胃镜、手术胃镜和超声胃镜，目标为食管黏膜下肿瘤、食管黏膜病变和胃炎，并采用患者分组五折划分。通过诊断结论规则匹配生成的初始检查级标签由两名分别来自两家参与医院、具有 10 年胃肠内镜临床经验的主任医师独立审核。主要多模态输出在四个数据集上的宏平均 F1 分别达到 0.9149、0.8877、0.8456 和 0.9064，优于所有比较的纯图像、纯文本和多模态方法。图像预算和位置消融支持 APro-CoPE 的有序序列建模作用；四个主要组件的 16 种组合消融实验支持超图推理、标签查询 cross-attention 和门控融合的联合贡献；多模态分支监督与知识蒸馏则改善了辅助纯图像输出。上述结果表明，我们提出的检查级多模态多标签框架能够在无需逐图标注的条件下，联合建模一次检查中的多张图像、配套所见描述与共存疾病，并在所见描述文本不可用时保留纯图像预测能力。
中文翻译：关键词：胃镜检查，检查级多标签分类，多实例学习，多模态学习，采集感知位置编码，知识蒸馏

Introduction

Upper gastrointestinal endoscopy is widely used to examine abnormalities of the esophagus and stomach. A gastroscopy examination contains a sequence of images acquired from different anatomical sites and viewing angles. A focal lesion may appear in only a few images, whereas diffuse inflammation may be visible across multiple images. In this study, we focus on three diseases that can be identified during upper gastrointestinal endoscopy: esophageal submucosal tumor, esophageal mucosal lesion, and gastritis. Esophageal submucosal tumors arise beneath the mucosal surface and often appear as smooth, localized elevations with relatively intact overlying mucosa. Esophageal mucosal lesions involve visible abnormalities of the mucosal surface, such as changes in color, erosion, or irregular morphology. Gastritis refers to inflammation of the gastric mucosa and may appear as erythema, edema, erosion, or other diffuse mucosal changes.
中文翻译：上消化道内镜被广泛用于检查食管和胃部异常。一次胃镜检查包含从不同解剖部位和观察角度采集的一系列图像。局灶性病变可能只出现在少数图像中，而弥漫性炎症可能出现在多张图像中。本文关注三个在上消化道内镜检查中可以发现的疾病：食管黏膜下肿瘤、食管粘膜病变和胃炎。食管黏膜下肿瘤起源于黏膜表面下方，在内镜下通常表现为表面黏膜相对完整的光滑局灶性隆起。食管粘膜病变是食管粘膜表面的可见异常，可表现为色泽改变、糜烂或形态不规则。胃炎是胃黏膜的炎症改变，可表现为充血、红斑、水肿、糜烂或其他弥漫性黏膜异常。

Most publicly available gastroscopy datasets are constructed at the image level and contain selected lesion images paired with category labels. Creating these datasets requires clinicians to inspect and annotate individual images, which is time-consuming and limits the scale of the data. Manual selection also favors clear and typical lesions, while a single image-level category may not accurately describe the transition or boundary between a lesion and normal tissue, leading to selection bias and label noise. Treating each image independently further ignores the acquisition order and contextual relationships among images from the same examination, a limitation not addressed by representative image-based classification and detection studies or by procedural quality-control systems. Public datasets and studies that retain complete examination sequences remain limited, although recent weakly supervised examination-level learning and systematically acquired short-sequence datasets have begun to explore this direction.
中文翻译：现有公开胃镜数据集大多以图像级形式构建，包含经过筛选的病灶图像及其类别标签。构建这类数据集需要医生逐张查看并标注图像，不仅耗时，也限制了数据集规模。人工筛选还容易偏向清晰、典型的病灶，而单一的图像级类别难以准确描述病灶与正常组织之间的过渡或边界区域，从而带来选择偏差和标签噪声。将每张图像独立处理还会忽略同一次检查中图像的采集顺序和上下文关系，而代表性的图像级分类、检测及流程质量控制研究尚未解决这一问题。尽管近期的检查级弱监督学习和系统采集的短序列数据集已开始探索这一方向，但保留完整检查序列的公开数据和相关研究仍然有限。

To address these data organization and annotation limitations, examination-level learning provides a practical alternative that assigns labels to the complete examination rather than to individual images, thereby reducing image-by-image annotation while preserving examination context. Multiple instance learning (MIL) is well suited to this setting because it treats each examination as an image bag and learns from bag-level labels. However, conventional MIL usually treats images as unordered instances and compresses them into a single bag representation. Most such approaches do not use positional encoding. Even when positional encoding is included, it usually assigns each sampled image a fixed position and therefore provides only a simple description of image order. Such a fixed representation cannot adapt positional relationships to image content and may therefore miss contextual changes across the examination. A shared bag representation is also limited for multi-label prediction because different diseases may be supported by different images. In addition, many multi-label methods predict each label independently or model only pairwise label relationships, which may not fully capture the higher-order relationships among several coexisting diseases. Image–text representation learning has demonstrated the value of textual semantics in both general and medical imaging domains. Nevertheless, existing examination-level MIL methods generally learn only from visual instances and do not fully exploit the semantic descriptions of lesion appearance and location in gastroscopic finding description text, limiting the use of complementary image–text information for examination-level disease representation. It also remains unclear whether knowledge learned jointly from paired image sequences and finding descriptions can improve an auxiliary image-only output when finding-description text is unavailable at inference.

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中文翻译：针对上述数据组织与标注局限，检查级学习提供了一种更符合临床归档形式的替代方案：它以完整检查为单位赋予标签，而非逐张图像标注，从而在保留检查上下文的同时降低逐图标注负担。多实例学习（MIL）与这一场景高度契合，因为它将每次检查视为一个图像 bag，并利用 bag 级标签进行学习。然而，传统 MIL 通常将图像视为无序实例，并将其压缩为一个整体 bag 表征。此类方法大多不使用位置编码；即使加入位置编码，通常也只是为每张采样图像分配一个固定位置，只能简单表示图像顺序。这种固定表示不能根据图像内容调整位置关系，因而难以充分表达完整检查中的上下文变化。共享的 bag 表征也不利于多标签预测，因为不同疾病可能由不同图像提供证据。此外，许多多标签方法独立预测各个标签，或只建模标签之间的两两关系，难以充分捕获多个共存疾病之间的高阶关系。图文表征学习已在通用和医学影像领域显示出文本语义的应用价值。然而，现有检查级 MIL 方法通常只从视觉实例中学习，没有充分利用胃镜所见描述文本中关于病灶形态和位置的语义信息，因而难以利用图像与文本的互补信息形成更完整的检查级疾病表征。此外，当推理阶段无法获得所见描述文本时，图像序列与所见描述联合学习得到的知识能否进一步改善辅助纯图像输出，仍缺少研究。

To address these gaps, we propose *AMEF-MIL* (**A**cquisition-aware **M**ultimodal **E**ndoscopic **F**inding-integrated **M**ultiple **I**nstance **L**earning) for examination-level multi-label classification of gastroscopy examinations. The framework integrates complete ordered image sequences with category-masked gastroscopic finding description text. Its acquisition-aware image encoder uses APro-CoPE to combine acquisition order with image content and provide more informative position representations. The primary multimodal branch performs label-wise cross-modal fusion using the image sequence and finding-description text, while an auxiliary image-only branch supports inference without finding-description text. We further investigate whether direct supervision of the multimodal branch and knowledge distillation from a pretrained, frozen multimodal teacher can transfer complementary multimodal knowledge to the image-only output. Both mechanisms are used only during training; all image-only configurations use endoscopic images alone at inference.

中文翻译：针对上述问题，本文提出用于胃镜检查级多标签分类的 *AMEF-MIL*（**A**cquisition-aware **M**ultimodal **E**ndoscopic **F**inding-integrated **M**ultiple **I**nstance **L**earning），即采集感知的多模态内镜所见融合多实例学习模型。该框架融合完整有序图像序列和经过类别名称掩码的胃镜所见描述文本，其采集感知图像编码器通过 APro-CoPE 结合图像采集顺序和图像内容，提供更有效的位置表示。作为主要输出，多模态分支利用图像序列与所见描述文本进行标签级跨模态融合；辅助纯图像分支则支持不使用所见描述文本的推理。本文进一步研究多模态分支直接监督，以及来自预训练且冻结的多模态教师的知识蒸馏，能否将互补的多模态知识迁移至纯图像输出。两种机制均仅在训练阶段使用，所有纯图像配置在推理阶段都只输入内镜图像。

The main contributions of this study are summarized as follows:

- We construct four examination-level multimodal gastroscopy datasets covering white-light endoscopy (WLE), chromoendoscopy, surgical gastroscopy, and endoscopic ultrasonography (EUS). Each dataset contains complete image sequences, examination-level multi-label annotations, and gastroscopic finding description text, supporting weakly supervised learning without image-by-image annotation.
- We propose APro-CoPE, which combines the original image acquisition order with image content to provide content-adaptive, acquisition-aware position representations for a complete examination.
- We develop multi-label attention MIL so that each disease can aggregate its own supporting image evidence, together with a label hypergraph reasoner that enables higher-order information exchange among coexisting diseases.
- We introduce label-query cross-attention and gated residual fusion to align each disease-specific visual representation with relevant token-level evidence from category-masked gastroscopic finding descriptions, thereby avoiding indiscriminate global image–text fusion.
- We develop a dual-branch output strategy centered on multimodal prediction from ordered images and finding descriptions while retaining an image-only output for text-free inference. We separately evaluate training-time multimodal branch supervision, knowledge distillation from a pretrained and frozen multimodal teacher, and their combination to transfer multimodal knowledge to the image-only branch.

中文翻译：本文的主要贡献总结如下：

- 本文构建四套检查级多模态胃镜数据集，覆盖白光胃镜（white-light endoscopy，WLE）、染色胃镜、手术胃镜和超声胃镜（endoscopic ultrasonography，EUS）。每套数据集均包含完整检查图像序列、检查级多标签标注以及胃镜所见描述文本，支持无需逐图标注的检查级弱监督学习。
- 本文提出 APro-CoPE，将图像的原始采集顺序与图像内容相结合，为完整检查提供内容自适应且具有采集感知能力的位置表征。
- 本文设计多标签注意力 MIL，使每种疾病能够聚合各自的支持图像证据，并通过标签超图推理器实现共存疾病之间的高阶信息交互。
- 本文引入标签查询 cross-attention 和门控残差融合机制，将每种疾病的视觉表征与经过类别名称掩码的胃镜所见描述中的相关 token 级证据对齐，从而避免不加区分的全局图文融合。
- 本文设计以有序图像和所见描述的多模态预测为主、同时保留纯图像输出的双分支策略，使模型能够在无文本条件下进行纯图像推理。本文分别评价训练阶段的多模态分支监督、来自预训练且冻结多模态教师的知识蒸馏及二者联合使用，以研究多模态知识向纯图像分支的迁移作用。

At the time this study was initiated, our review of the publicly available literature identified no previous study that jointly modeled multiple images acquired during a single examination and their paired finding descriptions for examination-level multi-label classification of coexisting diseases.

中文翻译：在本文研究开展阶段，据公开文献检索，尚未见将一次检查采集的多张图像与配套所见描述联合建模，用于检查级多标签共存疾病分类的研究。

Related Work

Endoscopic Datasets and Examination-Level Multi-Image Learning

Pogorelov et al. addressed the lack of standardized gastrointestinal endoscopy benchmarks by constructing Kvasir, which organizes labeled images into anatomical and pathological categories and supports reproducible image-level evaluation.[1] To improve dataset scale and diversity, Borgli et al. developed HyperKvasir with labeled images, unlabeled images, and videos, providing a broader resource for gastrointestinal image and video analysis.[2] Because performance differences among gastrointestinal classification methods were not yet well characterized, Jha et al. conducted a comprehensive comparative analysis and provided systematic evidence for selecting and evaluating image-level classifiers.[3] These studies established important public benchmarks, but their commonly used labeled data remain organized mainly at the image or video-category level.

Takiyama et al. addressed anatomical-site recognition by training a deep convolutional network to classify individual upper gastrointestinal endoscopy images, demonstrating that endoscopic anatomy can be recognized automatically.[4] Hirasawa et al. developed a convolutional neural network for detecting gastric cancer in selected endoscopic images, showing the potential of deep learning for lesion screening.[5] Luo et al. extended cancer detection to a real-time multicenter setting, demonstrating that an artificial intelligence system can support upper gastrointestinal cancer detection across clinical environments.[6] For procedural quality control, Wu et al. proposed WISENSE to identify blind spots during upper gastrointestinal endoscopy, establishing the value of acquisition-process information.[7] However, these methods focus on individual images, video frames, or procedural completeness rather than examination-level multi-label disease classification.

To reduce reliance on image-by-image annotation, Buendgens et al. linked routine endoscopy images with examination-level diagnoses and proposed a weakly supervised end-to-end framework, demonstrating that large clinical archives can support examination-level learning.[8] To preserve systematically acquired sequence information, Bravo et al. constructed GastroHUN with patient-linked images and short sequences collected under a standardized stomach screening protocol, providing a benchmark for anatomical image and sequence classification.[9] Nevertheless, datasets and methods for multi-label classification from complete ordered gastroscopy examinations remain limited. Our study therefore constructs four examination-level multimodal datasets and models the ordered image evidence within each complete examination.

中文翻译： Pogorelov 等人针对胃肠内镜领域缺少标准化公开基准的问题，构建 Kvasir 数据集，将标注图像划分为解剖和病理类别，为可重复的图像级评价提供了基础。为进一步提高数据规模和多样性，Borgli 等人开发 HyperKvasir，引入标注图像、未标注图像和视频，为胃肠内镜图像与视频分析提供了更广泛的数据资源。针对不同胃肠内镜分类方法性能差异尚不明确的问题，Jha 等人开展系统性比较分析，为图像级分类器的选择和评价提供了实验依据。这些研究建立了重要的公开基准，但其常用标注数据仍主要按照单张图像或视频类别组织。

Takiyama 等人针对内镜解剖部位自动识别问题，训练深度卷积网络对单张上消化道内镜图像进行分类，证明了自动识别内镜解剖结构的可行性。Hirasawa 等人针对胃癌筛查问题，开发卷积神经网络从筛选后的内镜图像中检测胃癌，显示了深度学习在病灶筛查中的潜力。Luo 等人进一步将癌症检测扩展到实时多中心环境，说明人工智能系统能够在不同临床场景中辅助上消化道癌检测。针对检查流程质量控制，Wu 等人提出 WISENSE 识别上消化道内镜检查中的盲区，证明了采集流程信息的应用价值。不过，这些方法主要面向单张图像、视频帧或检查完整性，而不是检查级多标签疾病分类。

为减少逐图标注依赖，Buendgens 等人将常规内镜图像与检查级诊断关联，提出弱监督端到端学习框架，证明了利用大规模临床归档开展检查级学习的可行性。为保留系统采集的序列信息，Bravo 等人构建 GastroHUN，提供按照标准化胃部筛查流程采集的患者关联图像和短序列，为解剖图像及序列分类建立了基准。然而，针对完整有序胃镜检查的多标签分类数据和方法仍然有限。因此，本文构建四套检查级多模态数据集，并对每次完整检查中的有序图像证据进行建模。

Multi-Label MIL and Higher-Order Label Reasoning

Ilse et al. addressed learning from bags without instance annotations by proposing attention-based MIL, which learns instance weights and produces an interpretable weighted bag representation.[10] To distinguish positive and negative instance evidence more effectively, Lu et al. proposed CLAM with attention-based aggregation and instance-level clustering constraints, improving weakly supervised whole-slide classification.[11] Li et al. developed DSMIL to combine instance and bag classifiers, allowing the most informative instances to guide bag-level representation learning.[12] Shao et al. proposed TransMIL to model correlations among instances with Transformer attention, strengthening contextual reasoning within large bags.[13] Zhang et al. developed DTFD-MIL to address the difficulty of learning from very large bags through double-tier feature distillation, improving the utilization of informative features.[14] Campanella et al. demonstrated that weakly supervised learning can scale to clinical-grade whole-slide classification using large routine pathology cohorts, establishing the practical value of bag-level supervision.[15] Although these approaches advance weakly supervised medical imaging, they generally generate a shared bag representation and therefore do not explicitly separate the image evidence required by different disease labels.

Ridnik et al. addressed severe positive–negative imbalance in multi-label learning by proposing asymmetric loss, which suppresses the contribution of easy negative samples and improves minority-label optimization.[16] To capture dependencies among labels, Chen et al. proposed ML-GCN, which constructs graph-convolutional classifiers from pairwise label co-occurrence and demonstrates the value of relational label modeling.[17] Because ordinary graphs cannot directly connect several nodes within one relation, Feng et al. proposed the hypergraph neural network, showing that hyperedges can represent higher-order relationships among multiple entities.[18] However, disease-specific image aggregation and higher-order label reasoning have not been jointly studied for complete gastroscopy examinations. Our method therefore combines multi-label attention MIL, which collects separate supporting images for each disease, with a label hypergraph reasoner that exchanges information among multiple coexisting findings.

中文翻译： Ilse 等人针对 bag 具有标签但实例缺少标注的问题，提出注意力 MIL，通过学习实例权重形成具有一定可解释性的加权 bag 表征。为更有效地区分阳性与阴性实例证据，Lu 等人提出 CLAM，将注意力聚合与实例级聚类约束结合，改善弱监督全切片分类。Li 等人开发 DSMIL，将实例分类器与 bag 分类器结合，使最具信息量的实例能够引导 bag 级表征学习。Shao 等人提出 TransMIL，利用 Transformer 注意力建模实例之间的相关性，增强大规模 bag 内部的上下文推理。Zhang 等人针对超大 bag 难以有效训练的问题提出 DTFD-MIL，通过双层特征蒸馏提高重要特征的利用效率。Campanella 等人利用大规模常规病理队列证明弱监督学习可以扩展到临床级全切片分类，体现了 bag 级监督的实际价值。尽管这些方法推动了弱监督医学影像分析，但其通常生成共享的 bag 表征，无法显式区分不同疾病标签所需要的图像证据。

Ridnik 等人针对多标签学习中严重的正负样本不平衡问题提出 asymmetric loss，通过抑制简单负样本的影响改善少数标签优化。为捕获标签依赖关系，Chen 等人提出 ML-GCN，根据标签两两共现关系构建图卷积分类器，证明了标签关系建模的价值。由于普通图难以在一个关系中直接连接多个节点，Feng 等人提出超图神经网络，说明超边能够表示多个实体之间的高阶关系。然而，疾病特异性图像聚合与高阶标签推理尚未在完整胃镜检查中得到联合研究。因此，本文将多标签注意力 MIL 与标签超图推理器结合，使每种疾病分别聚合自身的支持图像，并在多个共存发现之间进行信息交互。

Image–Report Multimodal Medical Learning

Li et al. addressed noisy image–text correspondence by proposing ALBEF, which aligns unimodal representations before multimodal fusion and uses momentum distillation to improve representation learning.[19] To reduce the effect of noisy web image–text pairs and unify vision–language tasks, Li et al. proposed BLIP with caption generation and filtering, improving the quality of multimodal pretraining data.[20] In medical imaging, Zhang et al. developed ConVIRT to learn visual representations from paired medical images and reports through contrastive learning, reducing dependence on manually annotated image labels.[21] Huang et al. proposed GLoRIA to address the mismatch between global reports and localized image findings through global–local representation alignment, improving label-efficient medical image recognition.[22] Zhou et al. developed REFERS to use cross-supervision between radiographs and free-text reports, demonstrating that report semantics can strengthen medical visual representation learning.[23] However, these methods mainly perform global or regional alignment and do not directly determine which text tokens support each disease label in a multi-label examination. Our primary multimodal branch therefore uses label-query cross-attention to retrieve disease-relevant tokens from category-masked gastroscopic finding descriptions and gated residual fusion to control their contribution.

Hinton et al. addressed the difficulty of deploying large or ensemble models by proposing knowledge distillation, which transfers softened teacher predictions to a smaller student and conveys information beyond hard labels.[24] Lopez-Paz et al. unified distillation with learning using privileged information, providing a framework in which additional information is available during training but not required at inference.[25] This idea is suitable for multimodal learning with paired images and finding descriptions, because a pretrained and frozen multimodal teacher can use both modalities to guide an image-only student during training, while the student uses only images at inference. Direct supervision of the multimodal branch provides a complementary route by optimizing the shared visual encoder with ground-truth labels. Prior image–report multimodal studies rarely separate these two transfer mechanisms. We therefore evaluate multimodal branch supervision, frozen-teacher knowledge distillation, and their combination for the auxiliary image-only output, while retaining the multimodal output as the primary prediction.

中文翻译： Li 等人针对图像与文本对应关系存在噪声的问题提出 ALBEF，在多模态融合前对齐单模态表征，并利用动量蒸馏改善表征学习。为降低网络图文对噪声并统一视觉语言任务，Li 等人提出 BLIP，通过图像描述生成与筛选提高多模态预训练数据质量。在医学影像领域，Zhang 等人开发 ConVIRT，通过配对医学图像和报告的对比学习获得视觉表征，降低对人工图像标签的依赖。Huang 等人针对全局报告与局部影像发现不匹配的问题提出 GLoRIA，通过全局—局部表征对齐改善标签高效的医学图像识别。Zhou 等人开发 REFERS，利用放射图像和自由文本报告之间的交叉监督，证明报告语义能够增强医学视觉表征。然而，这些方法主要进行全局或区域对齐，不能直接确定多标签检查中哪些文本 token 支持具体疾病。因此，本文的主要多模态分支使用标签查询 cross-attention 从经过类别名称掩码的胃镜所见描述中检索疾病相关 token，并通过门控残差融合控制其贡献。

Hinton 等人针对大型模型或集成模型难以部署的问题提出知识蒸馏，通过将教师的软化预测迁移给较小的学生模型，使学生获得硬标签之外的信息。Lopez-Paz 等人进一步统一知识蒸馏与特权信息学习，建立训练阶段可以使用额外信息、推理阶段不再需要该信息的学习框架。这一思想适用于图像与所见描述配对的多模态学习：预训练且冻结的多模态教师在训练阶段利用两种模态指导纯图像学生，而学生在推理阶段只使用图像。多模态分支直接监督则通过真实标签优化共享视觉编码器，形成另一条互补路径。现有图像–报告多模态研究较少区分这两种迁移机制。因此，本文分别评价多模态分支监督、冻结教师知识蒸馏及二者联合使用对辅助纯图像输出的作用，同时保留多模态输出作为主要预测。

Acquisition-Aware Position Modeling

Vaswani et al. addressed the lack of sequence-order information in attention models by introducing fixed positional encoding in the Transformer, enabling attention to distinguish elements at different positions.[26] Because fixed positions do not adapt to contextual importance, Golovneva et al. proposed CoPE to determine contextual positions from learned gates, making positional representation dependent on sequence content.[27] These methods established general solutions for sequence position modeling, but they were not designed for complete, non-causal gastroscopy examinations in which sampled images retain their original acquisition coordinates. We therefore propose APro-CoPE, which anchors positions to the original acquisition order and uses image content to construct monotonic contextual coordinates for acquisition-aware examination modeling.

中文翻译： Vaswani 等人针对注意力模型缺少序列顺序信息的问题，在 Transformer 中引入固定位置编码，使注意力能够区分不同位置的元素。由于固定位置不会随上下文重要性变化，Golovneva 等人提出 CoPE，通过学习得到的门控确定上下文位置，使位置表示能够依赖序列内容。这些方法为通用序列位置建模提供了基础，但并非针对完整、非因果的胃镜检查设计，也未考虑采样图像在原始检查中的采集坐标。因此，本文提出 APro-CoPE，以原始采集顺序为位置锚点，并根据图像内容构造保持单调的上下文坐标，实现采集感知建模。

Materials and Methods

Dataset construction and preprocessing

Study design and source data This retrospective observational study used integrated de-identified clinical data from Zhongshan Hospital, Fudan University, and Zhongshan Hospital (Xiamen), Fudan University. The two participating hospitals jointly provided four raw gastroscopy datasets organized by examination type: WLE, chromoendoscopy, surgical gastroscopy, and EUS. Each dataset contained endoscopic images, matched structured reports, and related fields from the hospitals' internal clinical databases. The study population consisted of patients whose examination records passed the data integrity and report-level screening procedures described below. Both participating hospitals agreed that the de-identified clinical archives could be used for this research. Because the data may contain sensitive clinical information and are subject to the participating hospitals' data governance requirements, the raw data are not publicly available; access to de-identified derived data may be considered upon reasonable request to the corresponding author and with permission from both hospitals.

中文翻译：本研究为一项回顾性观察性研究，使用整合自复旦大学附属中山医院和复旦大学附属中山医院厦门医院的去标识化临床数据。两家参与医院共同提供了按照检查类型组织的四个原始胃镜数据集，分别为 WLE、染色胃镜、手术胃镜和 EUS。每个数据集均包含内镜图像、与之匹配的结构化报告以及两院内部临床数据库中的相关字段。研究对象为检查记录通过下述数据完整性检查和报告级筛选流程的患者。两家参与医院均同意将去标识化临床归档数据用于本研究。由于数据可能包含敏感临床信息，并受两院数据管理规定约束，本研究所用原始数据不公开；经合理申请并获得两家医院许可后，可向通讯作者申请获取去标识化派生数据。

To clarify how routine clinical archive data were transformed into model-ready samples, we organized dataset construction into four stages: raw image–report collection, data cleaning and rule-based multi-label generation from diagnostic conclusions, independent clinical review, and examination-level multimodal sample construction. Each final sample contained a sampled image bag with a valid-image mask and the matched gastroscopic finding description text. Before entering the text encoder, direct target-category expressions in this text were masked to reduce label leakage. The complete workflow is summarized in Figure 1.

中文翻译：为了说明常规临床归档数据如何转化为模型可用样本，本文将数据集构建分为四个阶段：原始图像与报告收集、数据清洗和基于诊断结论的规则多标签生成、临床医师独立审核，以及检查级多模态样本构建。每个最终样本包含带有效图像掩码的采样图像 bag 和与之匹配的胃镜所见描述文本；文本进入模型前，对其中直接提示目标类别的表述进行掩码，以减少标签信息泄露。完整的数据处理流程见图 1。

Data cleaning, standardization, and clinical review of target labels Before constructing the task samples, we applied the same cleaning procedure to all four raw datasets. Together, they contained 2298 patients, 3530 examination directories, 215163 images, and 12444 examination reports. We first removed unreadable images and reports, and then excluded examination records with incomplete image–report correspondence. When multiple reports corresponded to the same examination, we merged the relevant report contents into a single examination-level record.

中文翻译：在构建任务样本前，我们对四个原始数据集采用相同的数据清洗流程。四个数据集合计包含 2298 名患者、3530 个检查目录、215163 张图像和 12444 份检查报告。我们首先删除无法读取的图像和报告，随后删除图像与报告对应关系不完整的检查记录。对于同一次检查对应多份报告的情况，我们将相关报告内容合并为一条检查级记录。

After data cleaning, records in each of the four datasets were standardized at the examination level. Each examination was treated as one independent record, and fields including report title, diagnostic conclusion, gastroscopic finding description, biopsy site, Helicobacter pylori status, and image count were formatted consistently. Based on these standardized records, we used rule-based text matching of the physician-written diagnostic conclusions to identify three non-mutually-exclusive initial target categories. The esophageal submucosal tumor label matched expressions related to esophageal submucosal tumor, submucosal elevation, and protruding lesion. The esophageal mucosal lesion label matched expressions related to esophageal mucosal lesion, esophageal mass, occupation, and neoplasm. The gastritis label matched chronic, active, inactive, atrophic, erosive, superficial, and bile reflux gastritis, as well as Kimura–Takemoto atrophy grades C1–O3. After label matching, records were excluded when the diagnostic conclusion was empty, no target label was identified, or only uncertain descriptions were present without satisfying the matching rules. Examinations matching at least one target label were retained, including those with coexisting findings. The physician-written diagnostic conclusions were used only for initial label generation and were not used as model input during training, validation, or testing.

中文翻译：完成数据清洗后，我们分别将四个数据集中的记录标准化为检查级结构化记录。每次检查作为一条独立记录，并统一格式化报告标题、诊断结论、胃镜所见描述、活检部位、幽门螺杆菌状态和图像数量等字段。在此基础上，我们对医生书写的诊断结论进行规则匹配，识别三个互不排斥的初始目标类别。食管黏膜下肿瘤标签匹配食管黏膜下肿瘤、黏膜下隆起和隆起性病变等相关表述；食管粘膜病变标签匹配食管粘膜病变、

Figure 1. Dataset construction and preprocessing pipeline. Raw image–report records were cleaned and standardized, initial examination-level labels were generated from diagnostic conclusions and independently reviewed by two chief physicians, direct target expressions were masked in the gastroscopic finding description text, and patient-grouped subsets were prepared for multimodal learning.

食管肿物、食管占位和食管新生物等相关表述；胃炎标签匹配慢性、活动性、非活动性、萎缩性、糜烂性、浅表性和胆汁反流性胃炎，以及 Kimura–Takemoto 分型中的 C1–O3 萎缩分级。完成标签匹配后，我们删除诊断结论为空、未匹配任何目标标签，或仅包含不确定性描述且不满足匹配规则的记录；凡命中至少一个目标标签的检查均予以保留，包括存在多个共存发现的检查。医生书写的诊断结论仅用于初始标签生成，在训练、验证和测试阶段均不作为模型输入。

The rule-initialized examination-level labels were subsequently reviewed independently by two chief physicians, one from each data-providing hospital, each with 10 years of clinical experience in gastrointestinal endoscopy. Each physician reviewed the complete image sequence and full report for every examination. The clinically reviewed labels were used for all model training and evaluation.

中文翻译：规则初始化的检查级标签随后由两名分别来自两家数据提供医院、具有 10 年胃肠内镜临床经验的主任医师独立审核。每名医师均审核每次检查的完整图像序列和完整报告。经临床审核的标签用于全部模型训练与评估。

After data cleaning, gastroscopy screening, label generation, and clinical review, 2861 eligible examinations with 171591 images were retained and organized into four examination-setting datasets. The WLE dataset contained 938 patients, 977 examinations, and 79508 images; the chromoendoscopy dataset contained 335 patients, 340 examinations, and 29439 images; the surgical gastroscopy dataset contained 950 patients, 965 examinations, and 43706 images; and the EUS dataset contained 572 patients, 579 examinations, and 18938 images. Notably, because some patients underwent multiple examination types, patient membership overlapped across these datasets. However, the four gastroscopy datasets differed fundamentally in imaging principles and image appearance, and each examination had independently documented clinical diagnostic conclusions and gastroscopic finding descriptions. Consequently, the limited patient overlap did not entail shared examination-level information, and its impact on cross-dataset comparisons was negligible.

中文翻译：经过数据清洗、胃镜筛选、标签生成和临床审核后，共保留 2861 次符合条件的检查和 171591 张图像，并组成四个检查场景数据集。其中，WLE 数据集包含 938 名患者、977 次检查和 79508 张图像；染色胃镜数据集包含 335 名患者、340 次检查和 29439 张图像；手术胃镜数据集包含 950 名患者、965 次检查和 43706 张图像；EUS 数据集包含 572 名患者、579 次检查和 18938 张图像。值得一提的是，由于部分患者接受了多种类型的检查，各数据集的患者存在交叉。然而，四个胃镜数据集在成像原理与图像表现形式上存在本质差异，且每次检查均具有独立记录的临床诊断结论和胃镜所见描述。因此，有限的患者交叉并不造成检查级信息共享，其对跨数据集比较造成的影响可以忽略不计。

All experiments were conducted independently on each of the four gastroscopy datasets using five-fold cross-validation. The folds were constructed at the patient level with stratification by multi-label combination, ensuring that all examinations from the same patient remained within the same fold. In each run, three folds were used for training, one for validation, and one for testing; across the five runs, each fold served as the test set once. The fold-specific numbers of training, validation, and test examinations are summarized in Table 1.

中文翻译：所有实验均在四个胃镜数据集上分别进行五折交叉验证。数据按照患者进行分组，并依据多标签组合完成分层，确保同一患者的全部检查始终位于同一折。在每轮实验中，三折用于训练，一折用于验证，另一折用于测试；五轮实验中，每一折均作为测试集一次。各折训练集、验证集和测试集的检查数量汇总于表 1。

Before model training, we quantified label co-occurrence separately in the four gastroscopy datasets. The original matrices in the upper row of Figure 2 reveal pronounced, dataset-specific class imbalance. Gastritis was predominant in the WLE dataset; both esophageal mucosal lesions and gastritis were frequent in the chromoendoscopy dataset; esophageal mucosal lesions were predominant in the surgical gastroscopy dataset; and esophageal submucosal tumors were predominant in the EUS dataset. We therefore applied multi-label minority oversampling exclusively to the training portion of each fold, while leaving the validation and test sets unchanged. For each gastroscopy dataset, the lower row of Figure 2 was obtained by pooling the five independently rebalanced fold-specific training sets and then calculating label co-occurrence over all resulting training exposures. Validation and test samples were not included in this pooled distribution.

中文翻译：模型训练前，我们分别统计四个胃镜数据集的标签共现关系。图 2上排的原始共现矩阵显示，各数据集均存在明显且具有数据集特异性的类别不平衡：WLE 数据集以胃炎为主，染色胃镜数据集中的食管黏膜病变和胃炎均较为常见，手术胃镜数据集以食管黏膜病变为主，EUS 数据集则以食管黏膜下肿瘤为主。因此，我们仅对每折训练集实施多标签少数类过采样，验证集和测试集保持不变。对于每个胃镜数据集，图 2下排先汇总五个折中分别完成重平衡的训练集，再基于汇总后的全部训练曝光计算标签共现比例；验证集和测试集不参与该汇总。

Across the eight matrices, direct co-occurrence between the two esophageal labels remained sparse, whereas their co-occurrence with gastritis varied substantially among the four gastroscopy datasets. Rebalancing reduced marginal class imbalance and increased training exposure to minority labels and their associated co-positive patterns. Together, these matrices distinguish the original label structure from the training-time exposure distribution and provide empirical motivation for explicitly modeling higher-order label relationships.

Table 1. Fold-specific examination counts for the four gastroscopy datasets. Each cell is formatted as training/validation/test. Counts are reported before training-set oversampling; validation and test sets were not rebalanced.

Fold	WLE	Chromoendoscopy	Surgical gastroscopy	EUS
Fold 1	588/194/195	207/67/66	579/193/193	346/117/116
Fold 2	582/201/194	205/68/67	579/193/193	346/116/117
Fold 3	582/194/201	204/68/68	579/193/193	347/116/116
Fold 4	590/193/194	201/71/68	579/193/193	349/114/116
Fold 5	589/195/193	203/66/71	579/193/193	349/116/114

中文翻译：综合这八个共现矩阵，两个食管标签之间的直接共现始终较少，而它们与胃炎的共现关系在四个胃镜数据集之间存在明显差异。重平衡减轻了边际类别不平衡，并增加了少数标签及其相关共同阳性模式的训练曝光。这八个矩阵共同区分了原始标签结构与训练阶段的曝光分布，并为显式建模高阶标签关系提供了经验依据。

Examination-level multimodal sample construction Each eligible examination was represented by its complete ordered image sequence, a three-dimensional multi-label target, and the gastroscopic finding description text. For model preparation, at most $T = 64$ images were uniformly sampled from each examination while preserving their original order. Each image was resized to 224×224 pixels. A binary mask $M_i = \{m_{i1}, \ldots, m_{iT}\}$ distinguished valid images from padded positions. The matched gastroscopic finding description text was retained as the text input; its category masking and encoding procedures are described in the text encoding section.

中文翻译：每次符合条件的检查由完整有序图像序列、三维多标签目标和胃镜所见描述文本表示。模型准备阶段从每次检查中最多均匀采样 $T = 64$ 张图像，并保持其原始顺序。每张图像被缩放至 224×224 像素。二值掩码 $M_i = \{m_{i1}, \ldots, m_{iT}\}$ 用于区分有效图像和填充位置。与检查匹配的胃镜所见描述文本作为文本输入，其类别掩码和编码过程将在文本编码部分中说明。

Overview of AMEF-MIL

As illustrated in Figure 3, AMEF-MIL organizes each examination as an ordered image bag, a valid-image mask, a finding description, and an examination-level multi-label target. The image branch applies a shared ConvNeXt-Tiny backbone and projection layer to obtain instance features. APro-CoPE preserves the original acquisition coordinates, constructs content-adaptive contextual coordinates, and delivers absolute and relative position signals to the bidirectional Transformer. Multi-label attention MIL then produces disease-specific visual representations, which the label hypergraph reasoner refines through higher-order interactions among coexisting findings. In parallel, category masking and a multi-scale TextCNN encode the finding description into token-level features. Each refined visual label representation retrieves disease-relevant textual evidence through label-wise cross-attention, and the gated residual module integrates the retrieved representation with its corresponding visual representation. The fused representations drive the primary multimodal branch, while the hypergraph-refined visual representations drive the auxiliary image-only branch. During training, ground-truth supervision jointly optimizes the two branches, and the pretrained multimodal teacher supplies prediction-level knowledge to the image-only branch. Figure 4 further presents the mathematical construction of APro-CoPE.

中文翻译：如图 3所示，AMEF-MIL 将每次检查组织为有序图像 bag、有效图像掩码、所见描述和检查级多标签目标。图像分支通过共享 ConvNeXt-Tiny 骨干网络和投影层提取实例特征。APro-CoPE 保留原始采集坐标，构建内容自适应上下文坐标，并向双向 Transformer 传递绝对与相对位置关系。多标签注意力 MIL 进一步生成疾病特异性视觉表征，标签超图推理器通过共存发现之间的高阶交互对其进行细化。文本分支通过类别名称掩码和多尺度 TextCNN 将所见描述编码为 token 级特征。每个经过细化的视觉标签表征通过标签查询 cross-attention 检索疾病相关文本证据，门控残差模块再将检索表征与对应视觉表征进行融合。融合表征驱动主要多模态分支，超图细化后的视觉表征驱动辅助纯图像分支。训练阶段的真实标签监督共同优化两个分支，预训练多模态教师则向纯图像分支提供预测级知识。图 4进一步展示 APro-CoPE 的数学构造。

Image Encoding

For the i-th gastroscopy examination, let $X_i = \{I_{i1}, I_{i2}, \ldots, I_{iN_i}\}$ denote its variable-length image bag, M_i the valid-image mask, and $y_i \in \{0, 1\}^L$ the examination-level target, where $L = 3$. The three dimensions correspond to esophageal submucosal tumor, esophageal mucosal lesion, and gastritis, and more than one label may be positive in the same examination. The image encoder maps X_i and M_i to label-specific visual representations that are subsequently used by both prediction branches.

中文翻译：对于第 i 次胃镜检查，令 $X_i = \{I_{i1}, I_{i2}, \ldots, I_{iN_i}\}$ 表示变长图像 bag，M_i 表示有效图像掩码，$y_i \in \{0, 1\}^L$ 表示检查级目标，其中 $L = 3$。三个维度分别对应食管黏膜下肿瘤、食管粘膜病变和胃炎，同一次检查可以同时具有多个阳性标签。图像编码器根据 X_i 和 M_i 生成标签特异性视觉表征，随后由两个预测分支共同使用。

Image encoding and APro-CoPE procedural context modeling Each sampled image was encoded by a shared ConvNeXt-Tiny backbone followed by a projection head, producing a $d = 512$ dimensional instance feature x_{it}.[28] A two-layer bidirectional Transformer supplied the intra-examination context backbone. Compared with conventional position encoding based on sampled-slot indices, we developed a more innovative Acquisition-Anchored Procedural Contextual Position Encoding (APro-CoPE). It was motivated by the context-dependent position increments of CoPE, but was reformulated for a non-causal clinical acquisition sequence with explicit acquisition anchors, monotonicity,

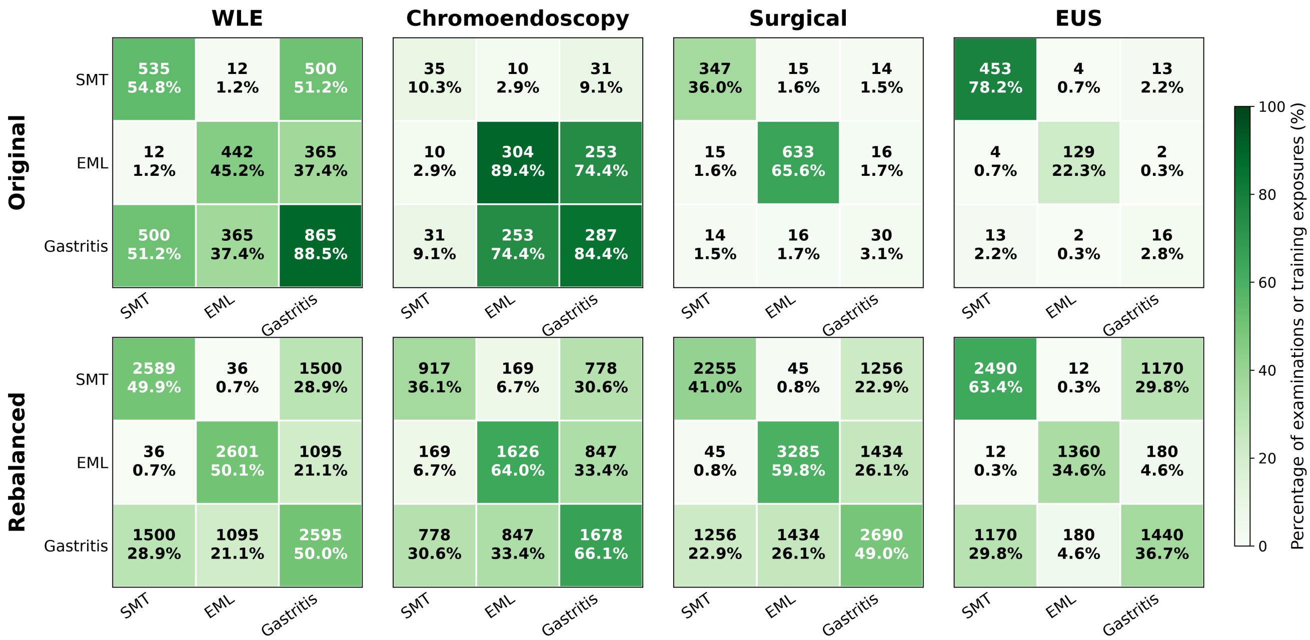

Figure 2. Original and rebalanced label co-occurrence matrices for the four gastroscopy datasets. Columns show the WLE, chromoendoscopy, surgical gastroscopy, and EUS distributions. The upper row presents the original examination distributions. The lower row presents the cumulative distribution obtained by pooling all five independently rebalanced training sets within each dataset; validation and test sets were not rebalanced or included. Each cell reports the corresponding count and percentage. Diagonal cells denote marginal positive frequencies for individual labels, whereas off-diagonal cells denote pairwise co-positive frequencies. Percentages are calculated relative to the corresponding original dataset or pooled training exposures. SMT, esophageal submucosal tumor; EML, esophageal mucosal lesion.

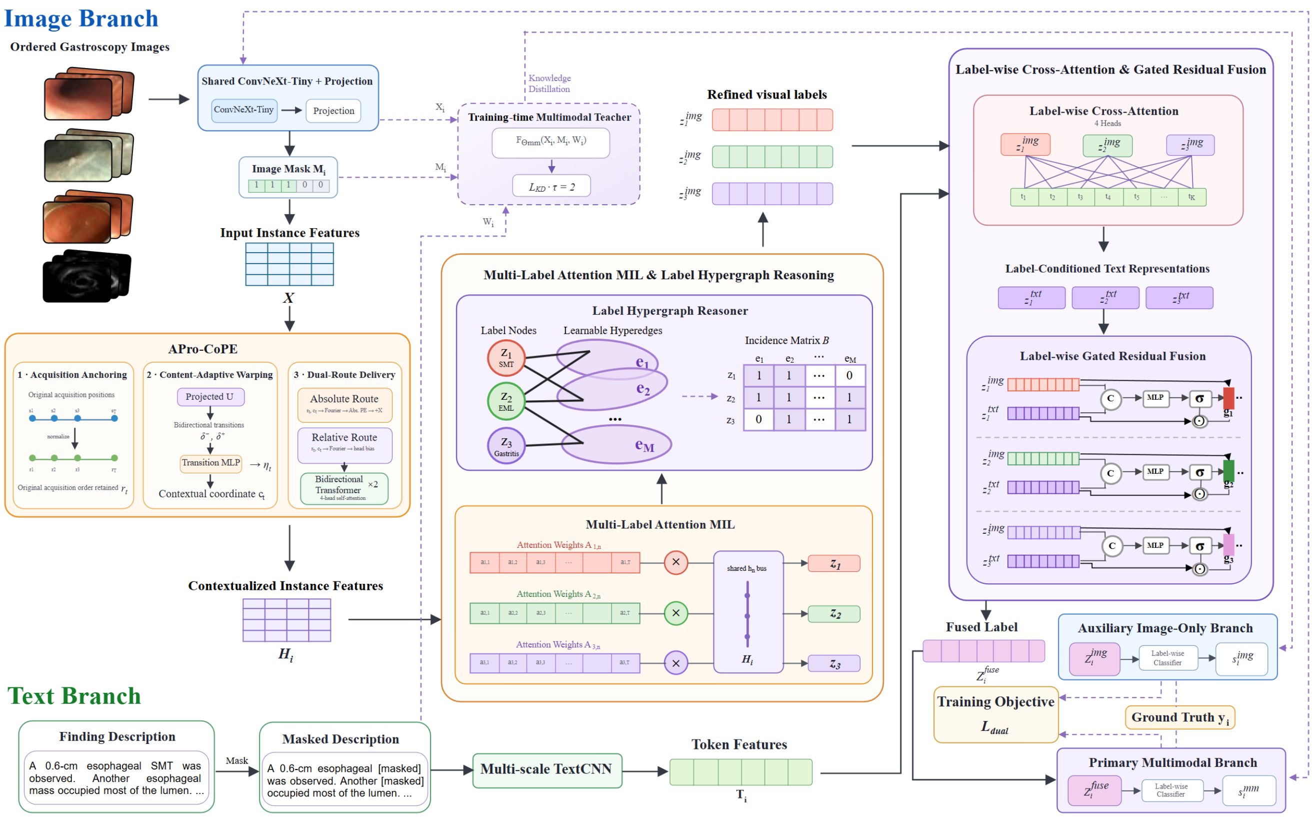

Figure 3. Overall architecture of AMEF-MIL. The image branch combines shared visual encoding, APro-CoPE-based bidirectional context modeling, multi-label attention MIL, and label hypergraph reasoning. The text branch encodes category-masked finding descriptions with a multi-scale TextCNN. Label-wise cross-attention and gated residual fusion generate the primary multimodal prediction, while the hypergraph-refined visual representations support the auxiliary image-only prediction and receive training-time knowledge from the pretrained multimodal teacher.

and endpoint conservation. Figure 4 summarizes its acquisition anchoring, contextual coordinate construction, and dual-route position delivery.

Let $s_{it} \in \{0, \ldots, N_i - 1\}$ be the original index of the t-th sampled image in an examination containing N_i acquired images. Its raw acquisition anchor is

$$r_{it} = \frac{s_{it}}{\max(N_i - 1, 1)}. \tag{1}$$

Consequently, uniform or non-uniform subsampling does not replace the original procedural coordinate with a new slot coordinate. We projected the visual feature as $u_{it} = \mathrm{LN}(W_p x_{it})$ and formed backward and forward transitions $\delta^-_{it} = u_{it} - u_{i,t-1}$ and $\delta^+_{it} = u_{i,t+1} - u_{it}$. A persistent transition score was then estimated from the current feature, the two signed transitions, their magnitudes, their element-wise agreement, and the raw acquisition gap:

$$\begin{aligned}\eta_{it} = \alpha \tanh\Big(\mathrm{MLP}\big([u_{it}, \delta^-_{it}, \delta^+_{it},\\ |\delta^-_{it}|, |\delta^+_{it}|, \delta^-_{it} \odot \delta^+_{it}, \Delta r_{it}]\big)\Big),\end{aligned} \tag{2}$$

where $\Delta r_{it} = r_{it} - r_{i,t-1}$ and $\alpha = 1.5$ bounds the deformation magnitude. The contextual gap and coordinate were defined as

$$g_{it} = \Delta r_{it} \exp(\eta_{it}), \tag{3}$$

$$\widetilde{g}_{it} = \frac{(r_{iT_i} - r_{i1}) g_{it}}{\sum_{k=2}^{T_i} g_{ik}}, \tag{4}$$

$$c_{i1} = r_{i1}, \qquad c_{it} = r_{i1} + \sum_{k=2}^{t} \widetilde{g}_{ik}, \tag{5}$$

where T_i is the number of valid sampled images. Because every warped gap is positive and the gaps are renormalized to the observed acquisition span, c_{it} remains monotonic and preserves both sequence endpoints.

APro-CoPE delivers position information through two complementary routes. The absolute route adds

$$\begin{aligned}p^{\mathrm{abs}}_{it} = f_{\mathrm{abs}}\big([\phi(r_{it}), \phi(c_{it}),\\ \phi(c_{it} - r_{it})]\big)\end{aligned} \tag{6}$$

to x_{it}, where $\phi(\cdot)$ is an eight-frequency continuous Fourier mapping. The relative route generates a head-specific attention bias

$$\begin{aligned}b^{(h)}_{itu} = 2 \tanh f^{(h)}_{\mathrm{rel}}\big([\phi(r_{it} - r_{iu}), \phi(c_{it} - c_{iu}),\\ \phi((c_{it} - c_{iu}) - (r_{it} - r_{iu}))]\big).\end{aligned} \tag{7}$$

The resulting features were processed by the two-layer, four-head bidirectional Transformer, with $b^{(h)}_{itu}$ added to the scaled dot-product attention logits. The valid-image mask excluded padded positions throughout, producing $H_i = \{h_{i1}, \ldots, h_{iT_i}\} \in \mathbb{R}^{T_i \times d}$.

中文翻译：每张采样图像由共享的 ConvNeXt-Tiny 骨干网络和投影头进行编码，得到 $d = 512$ 维实例特征 x_{it}。相比于传统的基于采样槽位编号的位置编码，本文提出更具创新性的采集锚定式流程上下文位置编码 APro-CoPE。该方法受 CoPE 上下文相关位置递增思想启发，但针对非因果临床采集序列重新设计，引入显式采集锚点、单调性约束和端点守恒。图 4概括了采集坐标锚定、上下文坐标构造以及双路径位置注入过程。

设 $s_{it} \in \{0, \ldots, N_i - 1\}$ 为第 t 张采样图像在包含 N_i 张原始图像的完整检查中的索引，其原始采集锚点定义为

$$r_{it} = \frac{s_{it}}{\max(N_i - 1, 1)}. \tag{8}$$

因此，无论采用均匀还是非均匀采样，原始流程坐标都不会被采样后的槽位坐标替代。首先将视觉特征投影为 $u_{it} = \mathrm{LN}(W_p x_{it})$，并构造后向转变 $\delta^-_{it} = u_{it} - u_{i,t-1}$ 和前向转变 $\delta^+_{it} = u_{i,t+1} - u_{it}$。随后根据当前特征、两个有符号转变、转变幅度、二者的逐元素一致性以及原始采集间隔估计持续转变分数：

$$\begin{aligned}\eta_{it} = \alpha \tanh\Big(\mathrm{MLP}\big([u_{it}, \delta^-_{it}, \delta^+_{it},\\ |\delta^-_{it}|, |\delta^+_{it}|, \delta^-_{it} \odot \delta^+_{it}, \Delta r_{it}]\big)\Big),\end{aligned} \tag{9}$$

其中 $\Delta r_{it} = r_{it} - r_{i,t-1}$，$\alpha = 1.5$ 用于限制坐标变形幅度。上下文间隔与上下文坐标定义为

$$g_{it} = \Delta r_{it} \exp(\eta_{it}), \tag{10}$$

$$\widetilde{g}_{it} = \frac{(r_{iT_i} - r_{i1}) g_{it}}{\sum_{k=2}^{T_i} g_{ik}}, \tag{11}$$

$$c_{i1} = r_{i1}, \qquad c_{it} = r_{i1} + \sum_{k=2}^{t} \widetilde{g}_{ik}, \tag{12}$$

其中 T_i 为有效采样图像数。由于变形后的每个间隔均为正值，且所有间隔会重新归一化到实际采集跨度，c_{it} 保持单调，并保留序列的起点和终点。

APro-CoPE 通过两条互补路径提供位置信息。绝对位置路径将

$$\begin{aligned}p^{\mathrm{abs}}_{it} = f_{\mathrm{abs}}\big([\phi(r_{it}), \phi(c_{it}),\\ \phi(c_{it} - r_{it})]\big)\end{aligned} \tag{13}$$

加入 x_{it}，其中 $\phi(\cdot)$ 表示包含 8 个频率的连续 Fourier 映射。相对位置路径生成注意力头特异的偏置

$$\begin{aligned}b^{(h)}_{itu} = 2 \tanh f^{(h)}_{\mathrm{rel}}\big([\phi(r_{it} - r_{iu}), \phi(c_{it} - c_{iu}),\\ \phi((c_{it} - c_{iu}) - (r_{it} - r_{iu}))]\big).\end{aligned} \tag{14}$$

最终特征输入包含两层和四个注意力头的双向 Transformer，并将 $b^{(h)}_{itu}$ 加入缩放点积注意力 logit。有效图像掩码在整个计算过程中排除填充位置，输出 $H_i = \{h_{i1}, \ldots, h_{iT_i}\} \in \mathbb{R}^{T_i \times d}$。

Multi-label attention MIL Different diseases may be supported by different images in the same examination. The model therefore learned one gated attention distribution for each label. For image n, the shared gated attention feature was

$$q_{in} = \tanh(V h_{in}) \odot \sigma(U h_{in}), \tag{15}$$

and the attention score and normalized weight for label ℓ were

$$\begin{aligned}e_{i\ell n} &= w_\ell^\top q_{in},\\ a_{i\ell n} &= \frac{m_{in} \exp(e_{i\ell n})}{\sum_{t=1}^{T} m_{it} \exp(e_{i\ell t}) + \epsilon}.\end{aligned} \tag{16}$$

The label-specific visual representation was obtained as

$$z_{i\ell} = \sum_{n=1}^{T} a_{i\ell n} h_{in}. \tag{17}$$

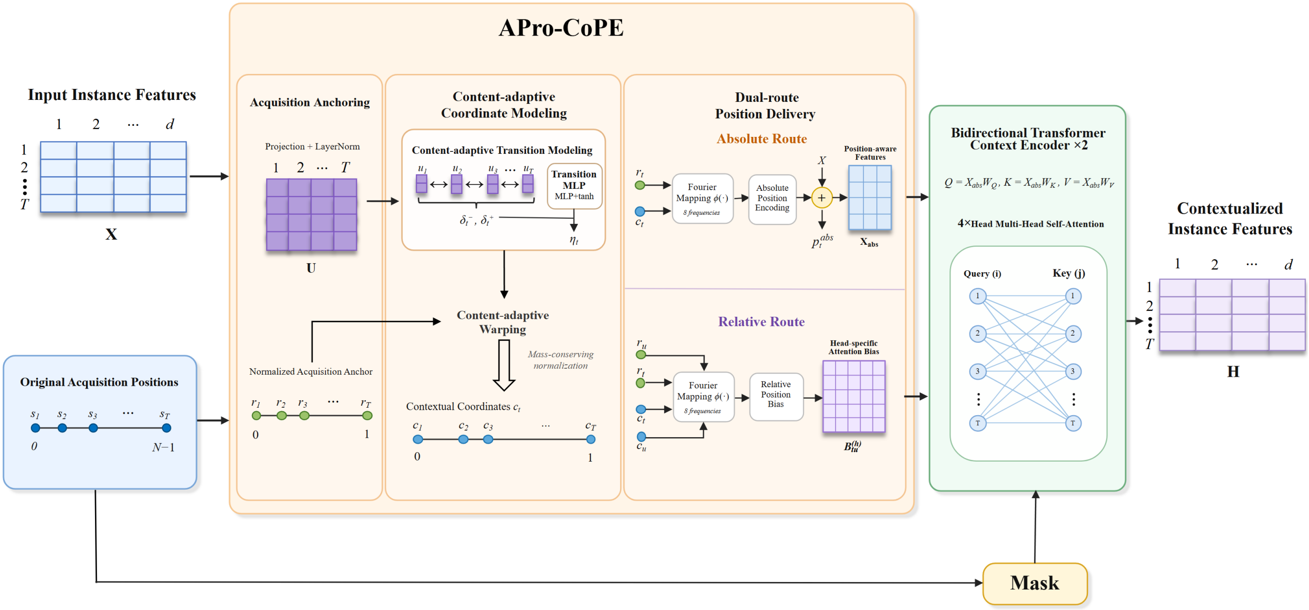

Figure 4. Structure of APro-CoPE. Original acquisition indices define normalized acquisition anchors. Bidirectional visual transitions are used to estimate bounded contextual deformation scores, and mass-conserving normalization constructs a monotonic contextual coordinate while preserving the observed sequence endpoints. The absolute route augments the instance features, whereas the relative route provides head-specific attention biases to the bidirectional Transformer.

This operation allows each disease to aggregate its own supporting images while ignoring invalid positions.

中文翻译：同一次检查中的不同疾病可能由不同图像提供证据，因此模型为每个标签分别学习门控注意力分布。对于图像 n，共享门控注意力特征为

$$q_{in} = \tanh(Vh_{in}) \odot \sigma(Uh_{in}), \tag{18}$$

标签 ℓ 对应的注意力得分和归一化权重为

$$e_{i\ell n} = w_\ell^\top q_{in},$$
$$a_{i\ell n} = \frac{m_{in}\exp(e_{i\ell n})}{\sum_{t=1}^{T} m_{it}\exp(e_{i\ell t}) + \epsilon}. \tag{19}$$

标签特异性视觉表征计算为

$$z_{i\ell} = \sum_{n=1}^{T} a_{i\ell n} h_{in}. \tag{20}$$

该操作使每种疾病能够聚合各自的支持图像，同时忽略无效位置。

Label hypergraph reasoning Let $Z_i = [z_{i1}, \ldots, z_{iL}]^\top \in \mathbb{R}^{L\times d}$ denote the label-specific visual feature matrix. To model higher-order relationships among coexisting diseases, the label hypergraph reasoner learned an incidence matrix $B \in \mathbb{R}^{L\times E}$ with $E = 2$ latent hyperedges. After normalization, information was propagated from label nodes to hyperedges and then back to label nodes:

$$U_i = S_{node\to edge}^\top \eta(Z_i), \tag{21}$$

$$\widetilde{Z}_i = S_{edge\to node} U_i, \tag{22}$$

$$Z_i^{img} = Z_i + \phi([Z_i; \widetilde{Z}_i]), \tag{23}$$

where $\eta(\cdot)$ and $\phi(\cdot)$ are learnable transformations. The resulting Z_i^{img} retains one hypergraph-refined visual representation for each disease and serves as the visual input to both prediction branches.

中文翻译：令 $Z_i = [z_{i1}, \ldots, z_{iL}]^\top \in \mathbb{R}^{L\times d}$ 表示标签特异性视觉特征矩阵。为建模多个共存疾病之间的高阶关系，标签超图推理器学习关联矩阵 $B \in \mathbb{R}^{L\times E}$，其中 $E = 2$ 表示两个潜在超边。归一化后，信息先从标签节点传播到超边，再从超边返回标签节点：

$$U_i = S_{node\to edge}^\top \eta(Z_i), \tag{24}$$

$$\widetilde{Z}_i = S_{edge\to node} U_i, \tag{25}$$

$$Z_i^{img} = Z_i + \phi([Z_i; \widetilde{Z}_i]), \tag{26}$$

其中 $\eta(\cdot)$ 和 $\phi(\cdot)$ 为可学习变换。所得 Z_i^{img} 为每种疾病保留一个经过超图细化的视觉表征，并作为两个预测分支共同使用的视觉输入。

Text Encoding

Finding-description selection and category masking The text encoder used the gastroscopic finding description text. This text describes lesion morphology, location, and surrounding mucosal changes, but it may also directly contain a target category name. To reduce direct label leakage, canonical disease names, abbreviations, and direct lexical variants associated with the three targets were replaced with a common mask token before tokenization. The masking lexicon covered direct category expressions such as esophageal submucosal tumor or SMT, esophageal mucosal lesion, and named forms of gastritis, while retaining anatomical locations and descriptive morphology. The same masking procedure was applied to all data subsets.

中文翻译：文本编码器使用胃镜所见描述文本。该文本记录病灶形态、位置和周围黏膜改变，但也可能直接包含目标类别名称。为减少直接标签泄露，模型在分词前将与三个目标相关的标准疾病名称、缩写和直接词汇变体替换为统一掩码 token。掩码词典覆盖食管黏膜下肿瘤或 SMT、食管粘膜病变以及不同命名形式的胃

炎等直接类别表述，同时保留解剖位置和描述性形态信息。所有数据子集使用相同的掩码过程。

TextCNN-based finding-description encoding After category masking, whitespace and punctuation were normalized, and each finding description W_i was tokenized into Chinese characters and alphanumeric fragments. The resulting tokens were mapped deterministically to a fixed vocabulary of 8192 indices and converted into 128-dimensional embeddings. Finding descriptions were truncated or padded to a maximum length of $K = 128$, with a binary text mask identifying valid token positions. Let $E_i \in \mathbb{R}^{K\times 128}$ denote the embedded token sequence. A multi-scale TextCNN applied parallel one-dimensional convolutions with kernel sizes $r \in \{2, 3, 4\}$ and same-length padding:

$$H_i^{(r)} = \mathrm{ReLU}(\mathrm{Conv1D}_r(E_i)), \qquad r \in \{2,3,4\}. \tag{27}$$

The multi-scale features were concatenated, normalized, and projected into the same $d = 512$ dimensional space as the visual features:

$$T_i = \mathrm{Dropout}\Big(\mathrm{GELU}\big(W_t\,\mathrm{LN}\big([H_i^{(2)}; H_i^{(3)}; H_i^{(4)}]\big) + b_t\big)\Big) \in \mathbb{R}^{K\times d}. \tag{28}$$

Padding positions in T_i were suppressed by the text mask, and the valid token-level representations were retained for label-query cross-attention. The embedding, convolutional, and projection parameters were optimized jointly with the complete model.

中文翻译：完成类别名称掩码后，首先规范化文本中的空格和标点，并将每份胃镜所见描述 W_i 切分为中文字符和字母数字片段。所得 token 以确定性方式映射至大小为 8192 的固定词表，并转换为 128 维嵌入。所见描述的最大长度设为 $K = 128$，超长序列被截断，较短序列使用填充补齐，同时采用二值文本掩码标记有效 token 位置。记嵌入后的 token 序列为 $E_i \in \mathbb{R}^{K\times 128}$，多尺度 TextCNN 使用卷积核大小 $r \in \{2, 3, 4\}$ 的并行一维卷积，并通过等长填充保持序列长度：

$$H_i^{(r)} = \mathrm{ReLU}(\mathrm{Conv1D}_r(E_i)), \qquad r \in \{2,3,4\}. \tag{29}$$

随后，将不同尺度的卷积特征进行拼接、归一化，并投影至与视觉特征相同的 $d = 512$ 维空间：

$$T_i = \mathrm{Dropout}\Big(\mathrm{GELU}\big(W_t\,\mathrm{LN}\big([H_i^{(2)}; H_i^{(3)}; H_i^{(4)}]\big) + b_t\big)\Big) \in \mathbb{R}^{K\times d}. \tag{30}$$

文本掩码用于抑制 T_i 中的填充位置，有效的 token 级表征被保留用于标签查询 cross-attention。嵌入层、卷积层和投影层参数与完整模型联合优化。

Cross-Modal Fusion

Label-query cross-attention and gated fusion The visual and textual representations encode complementary forms of evidence at different granularities: $Z_i^{img} \in \mathbb{R}^{L\times d}$ contains one examination-level representation for each disease, whereas $T_i \in \mathbb{R}^{K\times d}$ retains token-level finding-description information. Directly pooling T_i into a single vector would force all labels to share the same textual summary. We therefore used each hypergraph-refined visual label representation $z_{i\ell}^{img}$ as a query and the finding-description tokens as keys and values. For attention head $h \in \{1, \dots, H\}$, where $H = 4$ and $d_h = d/H$, the projected query, key, and value were

$$\begin{aligned} q_{i\ell}^{(h)} &= (z_{i\ell}^{img} + u_\ell)W_Q^{(h)}, \qquad k_{ik}^{(h)} = t_{ik}W_K^{(h)}, \\ v_{ik}^{(h)} &= t_{ik}W_V^{(h)}, \end{aligned} \tag{31}$$

where $u_\ell \in \mathbb{R}^d$ is a learnable label-identity embedding that preserves the distinction among disease queries during textual retrieval. Let $\mu_{ik} \in \{0, 1\}$ indicate whether the k-th textual position is valid. The label-to-token attention scores and normalized weights were calculated as

$$\begin{aligned} e_{i\ell k}^{(h)} &= \frac{q_{i\ell}^{(h)} k_{ik}^{(h)\top}}{\sqrt{d_h}}, \\ \alpha_{i\ell k}^{(h)} &= \frac{\mu_{ik}\exp(e_{i\ell k}^{(h)})}{\sum_{u=1}^{K}\mu_{iu}\exp(e_{i\ell u}^{(h)}) + \epsilon}. \end{aligned} \tag{32}$$

Thus, invalid padding positions did not participate in retrieval, and each disease label learned an independent attention distribution over the same finding description. The head-specific textual context and the resulting label-conditioned text representation were

$$\begin{aligned} o_{i\ell}^{(h)} &= \sum_{k=1}^{K}\alpha_{i\ell k}^{(h)} v_{ik}^{(h)}, \\ z_{i\ell}^{txt} &= \mathrm{Concat}_{h=1}^{H}\left(o_{i\ell}^{(h)}\right)W_O. \end{aligned} \tag{33}$$

Stacking the representations of all labels gives $Z_i^{txt} \in \mathbb{R}^{L\times d}$. This construction preserves label correspondence during fusion: the visual evidence for label ℓ retrieves textual evidence specifically for the same prediction rather than using a finding-description representation shared indiscriminately by all labels.

Because the relevance and reliability of the retrieved text can differ across diseases and examinations, a scalar gate was estimated separately for each label:

$$g_{i\ell} = a_i\,\sigma\Big(\rho([z_{i\ell}^{img}; z_{i\ell}^{txt}])\Big), \tag{34}$$

where $\rho(\cdot)$ is a two-layer multilayer perceptron and $a_i \in \{0, 1\}$ indicates whether the examination contains valid finding-description text. The fused representation was formed through gated residual injection:

$$z_{i\ell}^{fuse} = z_{i\ell}^{img} + g_{i\ell} z_{i\ell}^{txt}. \tag{35}$$

The residual form preserves the visual representation while allowing the model to introduce label-relevant semantic evidence with an examination- and label-dependent magnitude. When no valid finding-description text was available, $a_i = 0$ and $z_{i\ell}^{fuse} = z_{i\ell}^{img}$.

To explicitly constrain label–text correspondence, we further introduced a label-query discrimination loss. For a positive label ℓ and an absent label r, let $\tilde{s}_{i\ell\leftarrow r}^{mm}$ denote the counterfactual logit obtained by replacing $z_{i\ell}^{txt}$ with z_{ir}^{txt} while retaining the visual representation, gate, and classifier of label ℓ. The valid comparison set was $\mathcal{P}_i = \{(\ell, r) \mid a_i = 1,\ y_{i\ell} = 1,\ y_{ir} = 0,\ \ell \neq r\}$. The loss was defined as

$$\begin{aligned}\mathcal{L}_{LQD} = & -\frac{1}{\sum_i |\mathcal{P}_i|} \sum_i \sum_{(\ell,r)\in\mathcal{P}_i} \Big[\log \sigma(s^{mm}_{i\ell}) \\ & + \log\left(1 - \sigma(\widetilde{s}^{mm}_{i\ell\leftarrow r})\right) \Big].\end{aligned} \tag{36}$$

This objective encourages the text retrieved by the matched query to support its positive disease prediction while preventing text retrieved by an absent disease query from acting as interchangeable evidence. Pairs of coexisting positive labels were excluded so that the constraint did not suppress clinically valid co-occurrence information.

中文翻译：视觉表征与文本表征以不同粒度编码互补证据：$Z_i^{img} \in \mathbb{R}^{L\times d}$ 为每种疾病保留一个检查级表征，而 $T_i \in \mathbb{R}^{K\times d}$ 保留所见描述中的 token 级信息。若直接将 T_i 池化为单一向量，所有标签将被迫共享同一文本摘要。因此，本文将经过超图细化的视觉标签表征 $z_{i\ell}^{img}$ 作为 query，将所见描述 token 作为 key 和 value。对于注意力头 $h \in \{1, \ldots, H\}$，其中 $H = 4$ 且 $d_h = d/H$，query、key 和 value 的投影分别为

$$\begin{aligned} q^{(h)}_{i\ell} &= (z^{img}_{i\ell} + u_\ell)W^{(h)}_Q, \qquad k^{(h)}_{ik} = t_{ik}W^{(h)}_K, \\ v^{(h)}_{ik} &= t_{ik}W^{(h)}_V, \end{aligned} \tag{37}$$

其中 $u_\ell \in \mathbb{R}^d$ 为可学习的标签身份嵌入，用于在文本检索过程中保持不同疾病 query 之间的区分性。令 $\mu_{ik} \in \{0, 1\}$ 表示第 k 个文本位置是否有效，标签到 token 的注意力得分和归一化权重计算为

$$\begin{aligned} e^{(h)}_{i\ell k} &= \frac{q^{(h)}_{i\ell} k^{(h)\top}_{ik}}{\sqrt{d_h}}, \\ \alpha^{(h)}_{i\ell k} &= \frac{\mu_{ik} \exp(e^{(h)}_{i\ell k})}{\sum_{u=1}^K \mu_{iu} \exp(e^{(h)}_{i\ell u}) + \epsilon}. \end{aligned} \tag{38}$$

由此，填充位置不参与文本检索，每个疾病标签则在同一份所见描述上分别学习独立的注意力分布。各注意力头的文本上下文以及最终的标签条件文本表征为

$$\begin{aligned} o^{(h)}_{i\ell} &= \sum_{k=1}^K \alpha^{(h)}_{i\ell k} v^{(h)}_{ik}, \\ z^{txt}_{i\ell} &= \mathrm{Concat}_{h=1}^H \left(o^{(h)}_{i\ell} \right) W_O. \end{aligned} \tag{39}$$

将所有标签的表征堆叠后得到 $Z_i^{txt} \in \mathbb{R}^{L\times d}$。该设计在融合过程中保留标签对应关系，使标签 ℓ 的视觉证据能够检索服务于同一预测的文本证据，而不是对所有标签不加区分地使用同一个所见描述表征。

考虑到不同疾病和不同检查中所检索文本的相关性与可靠性并不相同，模型为每个标签分别估计一个标量门控系数：

$$g_{i\ell} = a_i\, \sigma\Big(\rho([z^{img}_{i\ell}; z^{txt}_{i\ell}])\Big), \tag{40}$$

其中 $\rho(\cdot)$ 为两层多层感知机，$a_i \in \{0, 1\}$ 表示该次检查是否包含有效所见描述文本。融合表征通过门控残差注入得到：

$$z^{fuse}_{i\ell} = z^{img}_{i\ell} + g_{i\ell} z^{txt}_{i\ell}. \tag{41}$$

残差形式保留原始视觉表征，同时允许模型根据具体检查和疾病标签，以自适应幅度引入相关语义证据。当不存在有效所见描述文本时，$a_i = 0$，此时 $z^{fuse}_{i\ell} = z^{img}_{i\ell}$。

为显式约束标签与文本之间的对应关系，本文进一步引入标签查询判别损失。对于阳性标签 ℓ 和未出现的标签 r，令 $\widetilde{s}^{mm}_{i\ell\leftarrow r}$ 表示将 $z^{txt}_{i\ell}$ 替换为 z^{txt}_{ir}、同时保持标签 ℓ 的视觉表征、门控和分类器不变时得到的反事实 logit。有效比较集合定义为 $\mathcal{P}_i = \{(\ell, r) \mid a_i = 1,\ y_{i\ell} = 1,\ y_{ir} = 0,\ \ell \neq r\}$，相应损失为

$$\begin{aligned}\mathcal{L}_{LQD} = & -\frac{1}{\sum_i |\mathcal{P}_i|} \sum_i \sum_{(\ell,r)\in\mathcal{P}_i} \Big[\log \sigma(s^{mm}_{i\ell}) \\ & + \log\left(1 - \sigma(\widetilde{s}^{mm}_{i\ell\leftarrow r})\right) \Big].\end{aligned} \tag{42}$$

该损失促使正确标签 query 检索的文本支持其阳性疾病预测，同时抑制未出现疾病 query 所检索的文本成为可互换证据。共存的阳性标签对不参与该约束，以保留临床上有效的疾病共存信息。

Prediction Branches

Primary multimodal branch Let $Z_i^{fuse} = [z^{fuse}_{i1}, \ldots, z^{fuse}_{iL}]^\top$ denote the label-wise fused representation. The primary multimodal branch applied a label-wise classifier to this representation:

$$s_i^{mm} = C_{mm}(Z_i^{fuse}). \tag{43}$$

Its parameters, together with the shared image encoder, text encoder, and cross-modal fusion modules, were optimized using examination-level asymmetric loss:

$$\mathcal{L}_{mm} = \mathcal{L}_{ASL}(s_i^{mm}, y_i) + \lambda_q \mathcal{L}_{LQD}. \tag{44}$$

Here, λ_q controls the label-query discrimination constraint. This branch constitutes the primary prediction pathway and uses both the endoscopic image sequence and the category-masked gastroscopic finding description.

For knowledge transfer to the image-only branch, a multimodal model with the same pathway was first pretrained using $\mathcal{L}_{mm}$. Its selected parameters were then fixed and denoted by $\overline{\Theta}_{mm}$. Given the same examination, the frozen teacher produced logits

$$\overline{s}_i^{mm} = F_{\overline{\Theta}_{mm}}(X_i, M_i, W_i), \tag{45}$$

where W_i denotes the category-masked finding description. The frozen teacher supplied soft targets but received no gradient during subsequent image-only training.

中文翻译：令 $Z_i^{fuse} = [z^{fuse}_{i1}, \ldots, z^{fuse}_{iL}]^\top$ 表示标签级融合表征。主要多模态分支将标签级分类器作用于该表征：

$$s_i^{mm} = C_{mm}(Z_i^{fuse}). \tag{46}$$

该分支参数与共享图像编码器、文本编码器及跨模态融合模块共同使用检查级非对称损失进行优化：

$$\mathcal{L}_{mm} = \mathcal{L}_{ASL}(s_i^{mm}, y_i) + \lambda_q \mathcal{L}_{LQD}. \tag{47}$$

其中 λ_q 用于控制标签查询判别约束的权重。该分支构成主要预测路径，同时使用内镜图像序列和经过类别名称掩码的胃镜所见描述。

为向纯图像分支迁移知识，首先使用 $\mathcal{L}_{mm}$ 预训练具有相同路径的多模态模型，随后固定所选模型参数，并记为 $\overline{\Theta}_{mm}$。对于同一次检查，冻结教师生成 logits：

$$\overline{s}_i^{mm} = F_{\overline{\Theta}_{mm}}(X_i, M_i, W_i), \tag{48}$$

其中 W_i 表示经过类别名称掩码的所见描述文本。冻结教师仅提供软目标，在后续纯图像训练过程中不接收梯度。

Image-only branch and multimodal knowledge transfer The auxiliary image-only branch predicted directly from the hypergraph-refined visual representations:

$$s_i^{img} = C_{img}(Z_i^{img}). \tag{49}$$

Its basic supervision used the same examination-level targets:

$$\mathcal{L}_{img} = \mathcal{L}_{ASL}(s_i^{img}, y_i). \tag{50}$$

When the trainable multimodal branch was supervised jointly, $\mathcal{L}_{mm}$ also updated the shared visual encoder and thereby provided an indirect representation-level learning signal to the image-only pathway.

Knowledge distillation provided a separate prediction-level transfer signal from the frozen multimodal teacher. With temperature τ, the teacher probability and multi-label distillation loss were defined as

$$\overline{p}_i^{mm} = \sigma\left(\frac{\overline{s}_i^{mm}}{\tau}\right), \tag{51}$$

$$\begin{aligned}\mathcal{L}_{KD} = \tau^2\,\mathrm{BCEWithLogits}\Bigg(&\frac{s_i^{img}}{\tau}, \\ &\mathrm{sg}(\overline{p}_i^{mm})\Bigg),\end{aligned} \tag{52}$$

where $\mathrm{sg}(\cdot)$ stops gradients through the teacher target. The joint objective was expressed as

$$\begin{aligned}\mathcal{L}_{dual} = \lambda_{img}\mathcal{L}_{img} &+ \lambda_{mm}\mathcal{L}_{mm} \\ &+ \lambda_{KD}\mathcal{L}_{KD},\end{aligned} \tag{53}$$

where λ_{img}, λ_{mm}, and λ_{KD} control the contributions of image-only supervision, multimodal branch supervision, and frozen-teacher distillation, respectively. In the combined configuration, they were set to 0.5, 1, and 1, respectively; λ_{mm} or λ_{KD} was set to 0 when the corresponding transfer mechanism was disabled. At image-only inference, only X_i, M_i, and C_{img} are required; the finding-description encoder, cross-modal fusion modules, and frozen teacher are not used.

中文翻译：辅助纯图像分支直接根据经过超图细化的视觉表征进行预测：

$$s_i^{img} = C_{img}(Z_i^{img}). \tag{54}$$

该分支使用相同的检查级目标进行基础监督：

$$\mathcal{L}_{img} = \mathcal{L}_{ASL}(s_i^{img}, y_i). \tag{55}$$

当可训练的多模态分支接受联合监督时，$\mathcal{L}_{mm}$ 还会更新共享视觉编码器，从而为纯图像路径提供间接的表征级学习信号。

知识蒸馏则由冻结的多模态教师提供独立的预测级迁移信号。温度为 τ 时，教师概率和多标签蒸馏损失分别定义为

$$\overline{p}_i^{mm} = \sigma\left(\frac{\overline{s}_i^{mm}}{\tau}\right), \tag{56}$$

$$\begin{aligned}\mathcal{L}_{KD} = \tau^2\,\mathrm{BCEWithLogits}\Bigg(&\frac{s_i^{img}}{\tau}, \\ &\mathrm{sg}(\overline{p}_i^{mm})\Bigg),\end{aligned} \tag{57}$$

其中 $\mathrm{sg}(\cdot)$ 表示教师目标不参与梯度反向传播。联合训练目标写为

$$\begin{aligned}\mathcal{L}_{dual} = \lambda_{img}\mathcal{L}_{img} &+ \lambda_{mm}\mathcal{L}_{mm} \\ &+ \lambda_{KD}\mathcal{L}_{KD},\end{aligned} \tag{58}$$

其中 λ_{img}、λ_{mm} 和 λ_{KD} 分别控制纯图像监督、多模态分支监督与冻结教师蒸馏的贡献。在联合配置中，三者依次设为 0.5、1 和 1；关闭相应迁移机制时，将 λ_{mm} 或 λ_{KD} 设为 0。纯图像推理时只需要 X_i、M_i 和 C_{img}，不使用所见描述编码器、跨模态融合模块或冻结教师。

Experiments

Experimental Settings

All experiments followed the patient-grouped five-fold protocol described above and used random seed 2026. Models were trained for at most 30 epochs with AdamW, an initial learning rate of 2×10^{-5}, a weight decay of 0.02, and an examination-level batch size of 1. Gradients were accumulated over four examinations. The learning rate was linearly warmed up during the first 20% of optimization steps and then decayed following a cosine schedule. Automatic mixed precision was enabled, dropout was set to 0.2, and asymmetric loss was used for multi-label supervision. The experiments were executed as independent fold-level jobs on six NVIDIA GeForce RTX 4090 GPUs.

For image encoding, input images were resized to 224×224 pixels and processed by a pretrained ConvNeXt-Tiny with its first stage frozen. Unless otherwise specified, up to 64 images were uniformly sampled from each examination, and the projected feature dimension was 512. The context encoder contained two bidirectional Transformer layers with four attention heads, and the label hypergraph contained two latent hyperedges. Full APro-CoPE used a 64-dimensional transition space, eight Fourier frequencies, $\alpha = 1.5$, mass conservation, and both absolute and relative position routes. For text encoding, finding descriptions were limited to 128 tokens and represented using a vocabulary of 8192 indices, 128-dimensional embeddings, and TextCNN kernels of sizes 2, 3, and 4. Label-query cross-attention used four heads, followed by label-wise gated residual fusion, and the label-query discrimination weight was $\lambda_q = 0.01$. The primary multimodal output used the endoscopic image sequence and category-masked gastroscopic finding description. In the image-only transfer experiments, λ_{img} was fixed at 0.5, λ_{mm} and λ_{KD} were independently set to 0 or 1 according to the training configuration, and the distillation temperature was $\tau = 2$.

Macro F1 was used as the primary evaluation and model-selection metric because it assigns equal importance to all three target labels under class imbalance. The checkpoint with the highest validation macro F1 was selected for each fold, and the final results are reported as the mean and standard deviation across the five patient-grouped folds. All experimental comparisons passed statistical significance testing.

中文翻译：所有实验均采用前文所述的患者分组五折协议，并使用随机种子 2026。模型采用 AdamW 训练最多 30 个 epoch，初始学习率为 2×10^{-5}，权重衰减为 0.02，检查级 batch size 为 1，并对连续 4 次检查累积梯

度。前 20% 优化步采用线性学习率预热，随后使用余弦衰减。训练过程中启用自动混合精度，dropout 设为 0.2，多标签监督采用非对称损失。所有实验以折为单位，在 6 张 NVIDIA GeForce RTX 4090 GPU 上独立运行。

图像端将输入图像缩放至 224×224 像素，并采用冻结第一阶段的预训练 ConvNeXt-Tiny 进行编码。除另有说明外，每次检查均匀采样最多 64 张图像，投影后的特征维度为 512。上下文编码器包含 2 层双向 Transformer 和 4 个注意力头，标签超图包含 2 个潜在超边。完整 APro-CoPE 采用 64 维转变空间、8 个 Fourier 频率、$\alpha = 1.5$、质量守恒以及绝对与相对双位置路由。文本端将所见描述长度限制为 128 个 token，采用大小为 8192 的词表、128 维嵌入以及卷积核大小为 2、3 和 4 的 TextCNN 进行编码。标签查询 cross-attention 包含 4 个注意力头，随后进行标签级门控残差融合，标签查询判别损失的权重设为 $\lambda_q = 0.01$。主要多模态输出同时使用内镜图像序列和经过类别名称掩码的胃镜所见描述。在纯图像迁移实验中，λ_{img} 固定为 0.5，λ_{mm} 和 λ_{KD} 根据训练配置分别设为 0 或 1，蒸馏温度设为 $\tau = 2$。

本文采用宏平均 F1 作为主要评价指标和模型选择指标，因为在类别不平衡条件下，该指标对三个目标标签赋予相同权重。每折选择验证集宏平均 F1 最高的检查点，最终结果报告患者分组五折的均值和标准差。所有实验比较均已通过统计显著性检验。

Overall Performance Comparison and Label-wise Analysis

Comparison with image-only, text-only, and multimodal models To evaluate overall examination-level classification performance, we compared AMEF-MIL with image-only MIL methods, text-only encoders, and multimodal classification models under the same patient-grouped five-fold protocol. The four multimodal comparison methods were reimplemented by adapting their fusion mechanisms to examination-level image bags and category-masked gastroscopic finding descriptions while retaining the common data splits and optimization settings.[29–32]

Table 2 presents the comparison results. AMEF-MIL achieved the highest mean macro F1 on all four datasets: 0.9149 on WLE, 0.8877 on chromoscopic gastroscopy, 0.8456 on surgical gastroscopy, and 0.9064 on EUS. In every dataset, AMEF-MIL outperformed the strongest image-only model and the strongest text-only model. This provides direct evidence that the model forms an effective multimodal representation by integrating the complementary discriminative information contributed by the ordered image sequence and the category-masked finding description. AMEF-MIL also outperformed all selected multimodal baselines across the four datasets, demonstrating the advantage of the complete pipeline—acquisition-aware image encoding, label hypergraph reasoning, disease-specific cross-attention, and gated residual fusion—for examination-level multi-label classification. This advantage was most pronounced on chromoscopic gastroscopy, where AMEF-MIL achieved both the highest mean macro F1 and the lowest fold-to-fold standard deviation among all compared methods.

中文翻译：为评价检查级分类的总体性能，本文在相同的患者分组五折协议下，将 AMEF-MIL 与纯图像 MIL 方法、纯文本编码器和多模态分类模型进行比较。对于 4 种多模态对比方法，本文将其融合机制适配至检查级图像 bag 与经过类别名称掩码的胃镜所见描述，同时保持统一的数据划分和优化设置。[29–32]

表 2给出了比较结果。AMEF-MIL 在四个数据集上均取得最高的平均宏平均 F1：WLE 为 0.9149、染色胃镜为 0.8877、手术胃镜为 0.8456、EUS 为 0.9064。在每个数据集上，AMEF-MIL 均优于表现最好的纯图像模型和纯文本模型。这一结果直接表明，模型能够整合有序图像序列与经过类别名称掩码的胃镜所见描述各自提供的判别信息，形成有效的跨模态互补表征，体现了 AMEF-MIL 的多模态建模能力。AMEF-MIL 在四个数据集上还均优于所选多模态基线，说明由采集感知图像编码、标签超图推理、疾病特异性 cross-attention 和门控残差融合构成的完整 pipeline 更适合检查级多标签分类任务。其中，染色胃镜上的优势最为突出，AMEF-MIL 同时取得全部对比方法中最高的平均宏平均 F1 和最小的折间标准差。

Out-of-fold label-wise confusion analysis Figure 5 further examines label-wise decisions by pooling predictions from the five mutually exclusive test folds, so each eligible examination contributes exactly once. Separate one-vs-rest 2×2 confusion matrices are reported because each examination may contain more than one positive label.

Positive-class sensitivity was high for most dataset–label combinations, including 97.9% for esophageal submucosal tumor and 97.7% for gastritis in WLE, 99.0% for esophageal mucosal lesion in chromoscopic gastroscopy, 94.5% for esophageal submucosal tumor in surgical gastroscopy, and 99.6% for esophageal submucosal tumor in EUS. The principal errors were false-positive predictions for labels with relatively few negative examinations, most visibly esophageal mucosal lesion and gastritis in chromoscopic gastroscopy and gastritis in WLE. These patterns complement macro F1 by showing how label prevalence and the selected decision thresholds affect the balance between sensitivity and specificity.

中文翻译：图 5通过汇总五个互斥测试折的预测进一步分析标签级判定，使每次符合条件的检查恰好参与一次折外评价。由于同一次检查可以具有多个阳性标签，本文分别报告一对其余 2×2 混淆矩阵。

多数“数据集–标签”组合具有较高的阳性类别敏感度，包括 WLE 中食管黏膜下肿瘤的 97.9% 和胃炎的 97.7%、染色胃镜中食管黏膜病变的 99.0%、手术胃镜中食管黏膜下肿瘤的 94.5%，以及 EUS 中食管黏膜下肿瘤的 99.6%。主要错误来自阴性检查相对较少标签的假阳性预测，其中染色胃镜中的食管黏膜病变和胃炎以及 WLE 中的胃炎较为明显。这些结果从标签层面补充了宏平均 F1，并显示标签患病率和所选判定阈值如何影响敏感度与特异度之间的平衡。

Analysis and Ablation of Model Components

Image budget and position encoding To examine whether increasing the number of sampled images allows the model to use a larger portion of each examination effectively, we evaluated six image budgets of 16, 32, 48, 64, 80, and 96 images per examination. At every image budget, we compared the same contextual encoder without position encoding, with sampled-slot position encoding, and with APro-CoPE, thereby separating the effect of examination coverage from the effect of position modeling.

The results in Table 3 show that increasing the image budget without position encoding produced progressively

Table 2. Comparison with image-only, text-only, and multimodal methods across the four gastroscopy datasets. Values are macro F1 reported as mean ± standard deviation over five patient-grouped folds. img. and txt. denote endoscopic image and category-masked gastroscopic finding description inputs, respectively.

Model	img.	txt.	WLE	Chromoscopic	Surgical	EUS
Attention MIL	✓	×	0.7919±0.0119	0.7391±0.0448	0.7018±0.1235	0.6268±0.0342
Mean pooling	✓	×	0.7931±0.0088	0.7408±0.0322	0.7173±0.0744	0.5910±0.0592
Transformer-context MIL	✓	×	0.8004±0.0160	0.7426±0.0194	0.7765±0.0886	0.5966±0.0612
Top-k MIL	✓	×	0.7862±0.0232	0.7511±0.0406	0.7698±0.0849	0.6037±0.0887
Max pooling	✓	×	0.7608±0.0034	0.6842±0.0261	0.6009±0.0919	0.5505±0.0756
TransMIL	✓	×	0.8021±0.0146	0.7217±0.0368	0.7516±0.0938	0.6045±0.0539
DSMIL	✓	×	0.7956±0.0159	0.7468±0.0448	0.6873±0.0852	0.5765±0.0765
DTFD-MIL	✓	×	0.7714±0.0141	0.7091±0.0306	0.7134±0.0853	0.5896±0.0419
CLAM-MB	✓	×	0.7741±0.0090	0.7137±0.0269	0.7094±0.0865	0.5883±0.0401
CLAM-SB	✓	×	0.7642±0.0040	0.6997±0.0234	0.6785±0.0479	0.5695±0.0379
Hashed mean	×	✓	0.8901±0.0184	0.8037±0.0689	0.8349±0.0187	0.8836±0.0688
Vocabulary attention	×	✓	0.8906±0.0124	0.8045±0.0708	0.8147±0.0487	0.8957±0.0542
TextCNN	×	✓	0.9004±0.0123	0.8652±0.0332	0.8356±0.0807	0.8982±0.0515
BiGRU	×	✓	0.8928±0.0191	0.8085±0.0818	0.8075±0.0342	0.9019±0.0531
Transformer	×	✓	0.8915±0.0163	0.8150±0.0281	0.8285±0.0174	0.8830±0.0604
MMFNet (2024) [29]†	✓	✓	0.9132±0.0099	0.8371±0.0647	0.8345±0.0844	0.8999±0.0731
RadFuse (2025) [32]†	✓	✓	0.9135±0.0083	0.7990±0.1013	0.8221±0.0671	0.8517±0.0712
SAIF (2025) [30]†	✓	✓	0.8413±0.0166	0.7201±0.0544	0.7482±0.0821	0.6266±0.0986
MMTF (2026) [31]†	✓	✓	0.9121±0.0118	0.7320±0.0415	0.7595±0.1371	0.7846±0.0974
AMEF-MIL (Ours)	✓	✓	**0.9149±0.0114**	**0.8877±0.0168**	**0.8456±0.0224**	**0.9064±0.0474**

†Reimplemented using the same examination-level inputs, category-masking procedure, patient-grouped folds, and optimization protocol as AMEF-MIL.

smaller gains, and the WLE performance even decreased locally at 48 and 80 images. This indicates that simply adding more images does not ensure that the contextual encoder can use the enlarged examination sequence effectively. Sampled-slot position encoding improved performance at nearly every matched budget and supported more consistent gains as additional images were introduced, particularly on chromoscopic gastroscopy, surgical gastroscopy, and EUS. APro-CoPE achieved the highest mean macro F1 in every matched dataset–budget comparison and was the only configuration whose mean performance increased monotonically from 16 to 96 images on all four datasets. The improvement was especially clear on chromoscopic gastroscopy: macro F1 increased from 0.8134 with 16 images to 0.8877 with 64 images, whereas the corresponding no-position values were 0.7478 and 0.7795. Beyond 64 images, APro-CoPE yielded only 0.16–0.52 additional percentage points across the four datasets, indicating that examination coverage approached a practical saturation point. We therefore used 64 images for the remaining ablations to retain most of the observed benefit while limiting computational cost.

中文翻译：为分析增加采样图像数量能否使模型有效利用更大范围的检查信息，本文评估每次检查采样 16、32、48、64、80 和 96 张图像的 6 种图像预算。在每种图像预算下，本文比较不使用位置编码、使用采样槽位位置编码和使用 APro-CoPE 的相同上下文编码器，从而区分检查覆盖范围与位置建模各自带来的影响。

表 3的结果表明，在不使用位置编码时，增加图像数量所带来的增益逐渐减小，WLE 在 48 张和 80 张图像时甚至出现局部性能下降。这说明，仅增加图像数量并不能保证上下文编码器有效利用扩展后的检查序列。加入采样槽位位置编码后，几乎所有匹配图像预算下的性能均得到改善，并且在染色胃镜、手术胃镜和 EUS 上随着图像数量增加呈现更稳定的提升。APro-CoPE 在每组匹配的“数据集–图像预算”比较中均取得最高的平均宏平均 F1，并且是唯一在四个数据集上均随图像数量从 16 张增加至 96 张而保持平均性能单调提升的配置。其中，染色胃镜上的提升尤为明显：APro-CoPE 的宏平均 F1 由 16 张图像时的 0.8134 提高至 64 张图像时的 0.8877，而不使用位置编码时相应结果仅由 0.7478 提高至 0.7795。超过 64 张图像后，APro-CoPE 在四个数据集上的额外增益仅为 0.16–0.52 个百分点，说明检查覆盖范围已接近实际性能饱和点。因此，后续消融实验统一采用 64 张图像，以在控制计算成本的同时保留大部分性能收益。

Component analysis of APro-CoPE To identify which design choices contribute to APro-CoPE, we fixed the image budget at 64 and used the same two-layer bidirectional Transformer for all configurations, changing only the position mechanism. The comparison includes a Transformer without position encoding, sampled-slot position encoding, bidirectional CoPE, acquisition-anchor position encoding, and component-level variants of APro-CoPE.

The results in Table 4 show that full APro-CoPE achieved the highest mean macro F1 on all four datasets. Compared with sampled-slot position encoding, it improved macro F1 by 1.70, 2.85, 2.13, and 0.65 percentage points on WLE, chromoscopic gastroscopy, surgical gastroscopy, and EUS, respectively. Replacing the persistent transition mechanism with pairwise transitions reduced performance on all four datasets, with the largest decrease on chromoscopic gastroscopy. Removing mass conservation produced decreases of 0.28–3.25 percentage points. Retaining only the absolute route also reduced performance, whereas retaining only the relative route caused larger decreases of 0.84, 11.66, 8.10, and 7.77 percentage points. Acquisition-anchor position encoding alone did not reproduce the performance of the

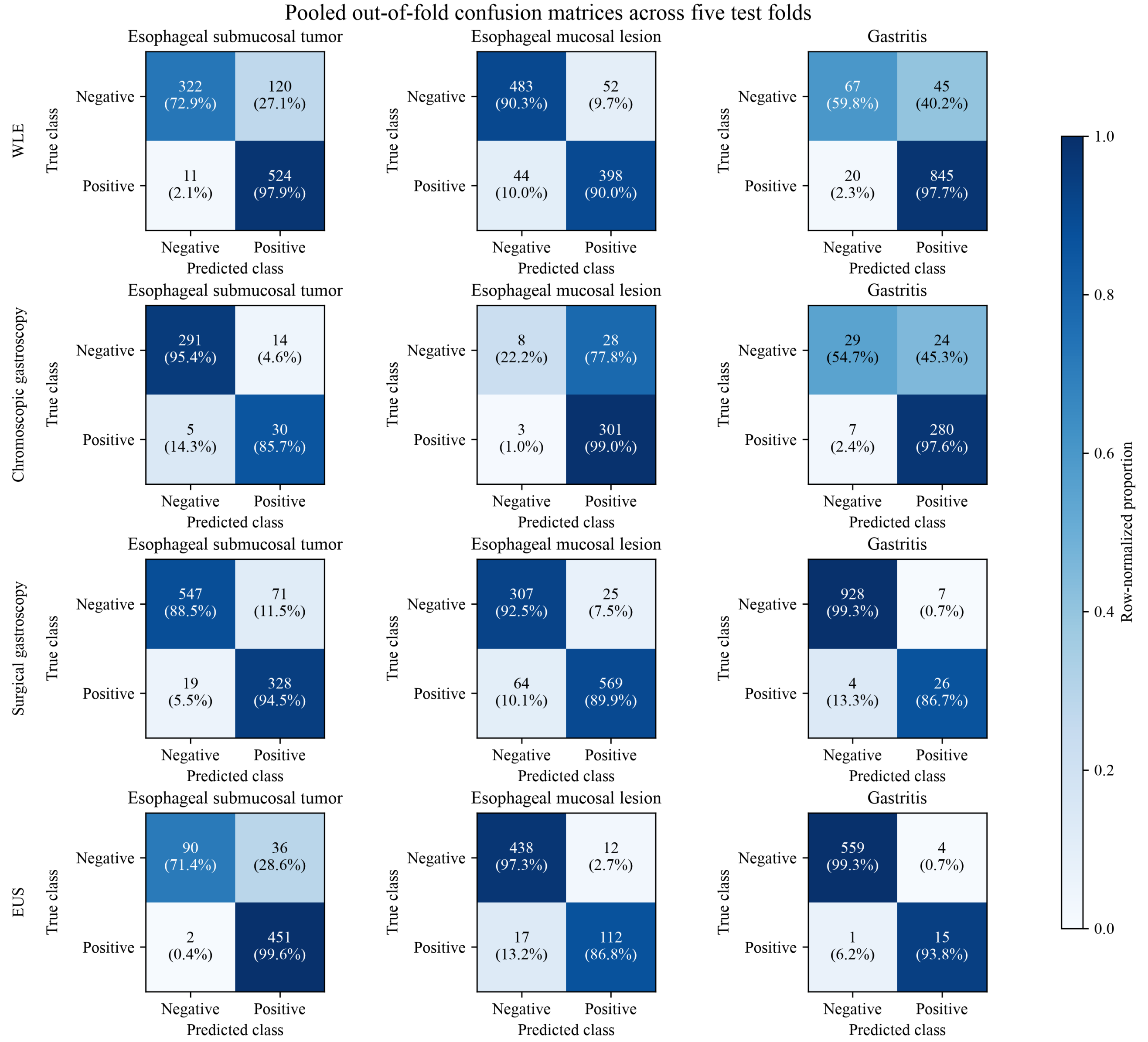

Figure 5. Pooled out-of-fold one-vs-rest confusion matrices of AMEF-MIL across the five patient-grouped test folds. Rows correspond to the WLE, chromoscopic gastroscopy, surgical gastroscopy, and EUS datasets, and columns correspond to the three target labels. Each cell reports the pooled count and the row-normalized percentage. Every examination appears in exactly one test fold.

full method. Taken together, these comparisons show that content-adaptive transition estimation, endpoint-preserving normalization, and dual-route position delivery add complementary information to the acquisition anchor; the absolute route is particularly important for maintaining stable performance across examination settings.

中文翻译：为确定 APro-CoPE 中不同设计的作用，本文将图像预算固定为 64 张，并在所有配置中使用相同的两层双向 Transformer，仅改变位置机制。对比配置包括不使用位置编码的 Transformer、采样槽位位置编码、双向 CoPE、采集锚点位置编码以及 APro-CoPE 的组件级变体。

表 4的结果表明，完整 APro-CoPE 在四个数据集上均取得最高的平均宏平均 F1。与采样槽位位置编码相比，其在 WLE、染色胃镜、手术胃镜和 EUS 上分别提高 1.70、2.85、2.13 和 0.65 个百分点。将持续转变机制替换为成对转变后，四个数据集的性能均有所下降，其中染色胃镜的下降最明显。移除质量守恒导致 0.28–3.25 个百分点的下降。仅保留绝对位置路径同样降低性能，而仅保留相对位置路径分别导致 0.84、11.66、8.10 和 7.77 个百分点的更大下降。仅使用采集锚点位置编码无法达到完整方法的性能。综合来看，内容自适应转变估计、端点保持归一化和双路径位置注入能够在采集锚点基础上提供互补信息，其中绝对位置路径对于在不同检查场景中维持稳定性能尤为重要。

Qualitative analysis of APro-CoPE under subsampling and full-sequence retention To qualitatively illustrate how APro-CoPE constructs acquisition-anchored, content-adaptive position relationships for both subsampled and fully retained examination sequences, we visualized representative cases under a maximum image budget of $T = 64$. An examination containing more than 64 images was represented by 64 uniformly subsampled images, with their original

Table 3. Effect of the sampled-image budget and position encoding across the four gastroscopy datasets. At each image budget, the same contextual encoder is evaluated without position encoding, with sampled-slot position encoding, and with APro-CoPE. Values are macro F1 reported as mean ± standard deviation over five patient-grouped folds. PE, position encoding.

Images	Method	WLE	Chromoscopic	Surgical	EUS
16	No PE	0.8673±0.0115	0.7478±0.0513	0.7824±0.0837	0.8443±0.0803
	Sampled-slot PE	0.8819±0.0132	0.7789±0.0683	0.8012±0.0919	0.8712±0.0663
	APro-CoPE (Ours)	**0.8986±0.0131**	**0.8134±0.0368**	**0.8168±0.0235**	**0.8879±0.0635**
32	No PE	0.8743±0.0078	0.7535±0.0341	0.7891±0.0676	0.8537±0.0758
	Sampled-slot PE	0.8840±0.0141	0.7978±0.0794	0.8125±0.0679	0.8836±0.0749
	APro-CoPE (Ours)	**0.9068±0.0133**	**0.8460±0.0297**	**0.8304±0.0215**	**0.8979±0.0564**
48	No PE	0.8720±0.0092	0.7635±0.0438	0.7898±0.0762	0.8594±0.0926
	Sampled-slot PE	0.8872±0.0148	0.8249±0.0690	0.8221±0.0865	0.8915±0.0672
	APro-CoPE (Ours)	**0.9119±0.0142**	**0.8732±0.0240**	**0.8408±0.0185**	**0.9030±0.0434**
64	No PE	0.8816±0.0086	0.7795±0.0351	0.8024±0.0882	0.8694±0.0974
	Sampled-slot PE	0.8979±0.0140	0.8592±0.0761	0.8243±0.0418	0.8999±0.0617
	APro-CoPE (Ours)	**0.9149±0.0114**	**0.8877±0.0168**	**0.8456±0.0224**	**0.9064±0.0474**
80	No PE	0.8814±0.0076	0.7818±0.0426	0.8041±0.0864	0.8715±0.1026
	Sampled-slot PE	0.8864±0.0106	0.8665±0.0508	0.8312±0.0369	0.9021±0.0598
	APro-CoPE (Ours)	**0.9158±0.0102**	**0.8906±0.0214**	**0.8480±0.0178**	**0.9089±0.0452**
96	No PE	0.8836±0.0091	0.7856±0.0397	0.8068±0.0828	0.8731±0.1034
	Sampled-slot PE	0.8930±0.0117	0.8706±0.0658	0.8345±0.0410	0.9042±0.0635
	APro-CoPE (Ours)	**0.9165±0.0090**	**0.8929±0.0177**	**0.8495±0.0197**	**0.9103±0.0442**

Table 4. Component analysis of APro-CoPE across the four gastroscopy datasets using 64 sampled images per examination. All variants use the same two-layer bidirectional Transformer, and only the position mechanism is changed. Values are macro F1 reported as mean ± standard deviation over five patient-grouped folds. w/o, without; PE, position encoding; Trans., transition mechanism; CG, content gate; PW, pairwise transition used in place of PT; PT, persistent transition; AA, acquisition anchor; MC, mass conservation; Abs., absolute route; Rel., relative route.

Position variant	Trans.	AA	MC	Abs.	Rel.	WLE	Chromoscopic	Surgical	EUS
Transformer w/o PE	--	--	--	--	--	0.8816±0.0086	0.7795±0.0351	0.8024±0.0882	0.8694±0.0974
Sampled-slot PE	--	--	--	✓	--	0.8979±0.0140	0.8592±0.0761	0.8243±0.0418	0.8999±0.0617
Bidirectional CoPE	CG	--	--	--	✓	0.9067±0.0059	0.7562±0.0357	0.7924±0.1004	0.8561±0.0996
Acquisition-anchor PE	--	✓	--	✓	--	0.9105±0.0039	0.8592±0.0586	0.8112±0.0736	0.8999±0.0731
APro-CoPE w/o Trans.	PW	✓	✓	✓	✓	0.9102±0.0070	0.8391±0.0759	0.8356±0.0807	0.8900±0.0572
APro-CoPE w/o MC	PT	✓	--	✓	✓	0.9121±0.0118	0.8552±0.0187	0.8131±0.0625	0.8974±0.0760
APro-CoPE w/o Rel.	PT	✓	✓	✓	--	0.9101±0.0057	0.8274±0.0811	0.8289±0.0681	0.8957±0.0658
APro-CoPE w/o Abs.	PT	✓	✓	--	✓	0.9065±0.0086	0.7711±0.0557	0.7646±0.1039	0.8287±0.0879
Full APro-CoPE (Ours)	PT	✓	✓	✓	✓	**0.9149±0.0114**	**0.8877±0.0168**	**0.8456±0.0224**	**0.9064±0.0474**

acquisition order and indices preserved; an examination containing at most 64 images retained its full image sequence. Figure 6 presents both input-construction settings using examples from the four gastroscopy datasets. For each example, the image strip presents the sequence supplied to the model, the orange curve measures the feature change between adjacent images, the blue dashed curve measures the relative interval change from original acquisition positions to uniformly spaced sampled slots, and the red curve measures the relative interval change from acquisition anchors to the contextual coordinates produced by APro-CoPE. The blue curve therefore characterizes the position change introduced during input construction, while the red curve characterizes the subsequent content-adaptive adjustment.

Uniform subsampling produced visible blue fluctuations because the selected images retained their positions within the complete acquisition sequence, whereas full-sequence retention yielded blue values close to zero through direct correspondence between image order and sampled slots. Against these acquisition anchors, the red curves formed distinct patterns for individual examinations under both input-construction settings, showing how APro-CoPE redistributed contextual intervals according to image content. The red deformation patterns followed the continuity of visual changes across neighboring images, reflecting content-adaptive interval adjustment driven by persistent image transitions. Particularly in the EUS examples, several isolated peaks in adjacent-image feature change coincided with relatively small overall red adjustments; by comparison, the WLE, chromoendoscopy, and surgical gastroscopy examples showed stronger adjustments within sequence segments containing sustained visual changes. Together, these patterns

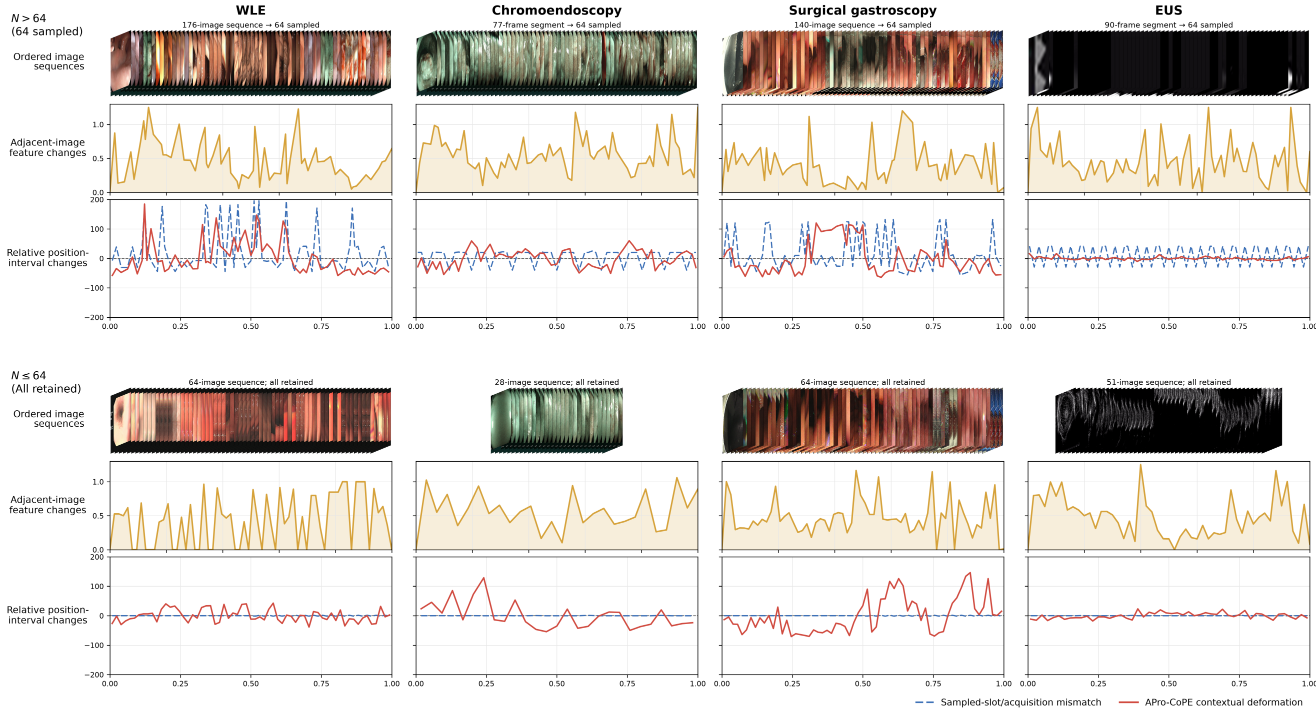

Figure 6. Qualitative visualization of APro-CoPE under subsampling and full-sequence retention. With a maximum image budget of $T = 64$, the upper group uniformly subsamples 64 images when $N > 64$, and the lower group retains the full sequence when $N \leq 64$.

illustrate examination-specific position modeling for both subsampled and fully retained inputs.

中文翻译：为定性展示 APro-CoPE 如何针对均匀子采样序列和完整保留序列构建采集锚定、内容自适应的位置关系，本文在最大图像预算 $T = 64$ 下选取代表性检查进行可视化。图像数量超过 64 张的检查采用 64 张图像的均匀子采样表示，同时保留入选图像的原始采集顺序和采集索引；图像数量至多为 64 张的检查保留完整图像序列。图 6通过四个胃镜数据集的示例展示这两种输入构建方式。对于每个示例，图像条带为实际输入模型的有序序列，橙色曲线表示相邻图像之间的特征变化，蓝色虚线表示从原始采集位置到均匀采样槽位的相对间隔变化，红色实线表示从采集锚点到 APro-CoPE 上下文坐标的相对间隔变化。因此，蓝色曲线刻画输入构建过程中产生的位置变化，红色曲线刻画模型随后进行的内容自适应调整。

均匀子采样保留了入选图像在完整采集序列中的位置，因此蓝色曲线呈现明显波动；完整序列保留使图像顺序与采样槽位直接对应，蓝色曲线相应接近于零。在这些采集锚点基础上，两种输入构建方式下的红色曲线均形成了面向具体检查的变化模式，展示 APro-CoPE 如何根据图像内容重新分配上下文位置间隔。红色变形模式与相邻图像之间视觉变化的连续性相对应，反映由持续图像转变驱动的内容自适应间隔调整。特别是在 EUS 示例中，若干孤立的相邻图像特征变化峰值与较小的整体红色调整同时出现；相比之下，WLE、染色胃镜和手术胃镜示例在包含持续视觉变化的序列区段呈现更明显的调整。上述结果体现 APro-CoPE 对均匀子采样输入和完整序列输入的检查特异性位置建模。

Effect of APro-CoPE on examination-level prediction quality Figure 7 compares the examination-level distributions of label-wise accuracy, true-class confidence, and Brier score between sampled-slot PE and full APro-CoPE under the 64-image setting. Label-wise accuracy is the proportion of the three binary labels classified correctly. True-class confidence is the mean probability assigned to the correct binary outcome, and the Brier score is the mean squared difference between the predicted probabilities and binary targets. Therefore, higher label-wise accuracy indicates more correct label decisions, higher true-class confidence indicates greater confidence in the correct outcomes, and a lower Brier score indicates smaller probability error; together, these values indicate better examination-level prediction quality.

Compared with sampled-slot PE, APro-CoPE achieved higher mean label-wise accuracy on WLE, chromoscopic gastroscopy, and EUS; only surgical gastroscopy showed a small decrease, from 0.9179 to 0.9160. Mean true-class confidence was also higher on WLE, chromoscopic gastroscopy, and surgical gastroscopy; only EUS showed a small decrease, from 0.9079 to 0.9025. The mean Brier score was lower with APro-CoPE on all four datasets. The largest reduction occurred on chromoscopic gastroscopy, from 0.0809 to 0.0730, followed by EUS, from 0.0463 to 0.0403. Thus, APro-CoPE improved label correctness and confidence in the correct outcomes on most datasets and reduced probability error on all four datasets, indicating an overall improvement in examination-level prediction quality. These distributional improvements also help explain the more pronounced advantage of APro-CoPE over sampled-slot PE on chromoscopic gastroscopy in the preceding experiments.

中文翻译：图 7比较了 64 张图像设置下采样槽位位置编码与完整 APro-CoPE 在标签级准确率、真实类别置信度和 Brier 分数上的检查级分布。标签级准确率表示三个二元标签中预测正确的比例；真实类别置信度表示模型赋予正确二元结果的平均概率；Brier 分数表示预测概率与真实标签之间的均方误差。因此，更高的标签级准

确率说明正确判定的标签更多，更高的真实类别置信度说明模型对正确结果的置信度更高，更低的 Brier 分数说明预测概率误差更小，三者共同反映更好的检查级预测质量。

与采样槽位位置编码相比，APro-CoPE 在 WLE、染色胃镜和 EUS 上取得了更高的平均标签级准确率，仅手术胃镜由 0.9179 小幅下降至 0.9160。WLE、染色胃镜和手术胃镜的平均真实类别置信度同样更高，仅 EUS 由 0.9079 小幅下降至 0.9025。APro-CoPE 在四个数据集上的平均 Brier 分数均更低，其中染色胃镜的降幅最大，由 0.0809 降至 0.0730；EUS 次之，由 0.0463 降至 0.0403。因此，APro-CoPE 在多数数据集上提高了标签判定正确性和对正确结果的置信度，并在全部四个数据集上减小了预测概率误差，表明其整体改善了检查级预测质量。这些分布层面的改善也有助于解释前述实验中 APro-CoPE 相较于采样槽位位置编码在染色胃镜数据集上表现出的更明显优势。

Ablation of multi-label attention MIL and label hypergraph reasoning To distinguish the contributions of label-specific image aggregation and higher-order label reasoning, we conducted the targeted ablation summarized in Table 5. Starting from shared-attention MIL without label reasoning, we introduced multi-label attention MIL and then compared no label reasoning, an ordinary learnable label graph, and the proposed label hypergraph.

Replacing shared-attention MIL with multi-label attention MIL consistently improved macro F1 by 0.56–0.61 percentage points across the four datasets. Adding the ordinary label graph produced further gains of 0.56–0.62 percentage points, while replacing it with the label hypergraph yielded larger additional gains of 1.86–3.59 percentage points. The model configuration combining multi-label attention with label hypergraph reasoning achieved the highest macro F1 on every dataset and improved over the shared-attention baseline by 3.09–4.76 percentage points, supporting the complementary roles of label-specific image aggregation and higher-order label reasoning.

中文翻译：为区分标签特异性图像聚合与高阶标签推理各自的贡献，本文开展了表 5所示的针对性消融实验。实验以不进行标签推理的共享注意力 MIL 为起点，引入多标签注意力 MIL，并进一步比较不使用标签推理、使用普通可学习标签图以及使用本文标签超图的配置。

将共享注意力 MIL 替换为多标签注意力 MIL 后，四个数据集上的宏平均 F1 均得到提升，增幅为 0.56–0.61 个百分点。进一步加入普通标签图带来 0.56–0.62 个百分点的增益，而将普通标签图替换为标签超图后又取得 1.86–3.59 个百分点的更大提升。采用多标签注意力与标签超图推理组合的模型配置在所有数据集上均取得最高宏平均 F1，相较共享注意力基线提高 3.09–4.76 个百分点，支持标签特异性图像聚合与高阶标签推理具有互补作用。

To determine whether the performance gains in Table 5 were accompanied by label-specific visual evidence allocation, we compared the overlap among label-wise image-evidence sets across the four gastroscopy datasets (Figure 8). Our model's multi-label attention module consistently remained among the methods with the strongest evidence separation, showing that it assigned differentiated supporting images to individual diseases across examination settings. This result provides a basis for further assessing whether the differentiated evidence contributes selectively to the corresponding label decisions.

中文翻译：为判断表 5中的性能增益是否伴随标签特异性的视觉证据分配，本文比较了四个胃镜数据集中不同标签图像证据集合的重叠情况（图 8）。我们模型的多标签注意力模块在各检查场景下均保持在证据分离能力最强的方法之列，表明该模块能够为不同疾病分配差异化的支持图像。该结果为进一步验证这些差异化证据是否选择性地参与对应标签决策提供了基础。

To further assess whether the separated evidence contributed selectively to the corresponding label decisions, we performed label-targeted deletion (Figure 9). Compared with the shared-attention control, our model's multi-label attention module exhibited stronger diagonal concentration in most examination settings. After removing the high-attention evidence associated with a given label, the resulting decision impact was concentrated more strongly on that label's own prediction, indicating that the module learned label-specific visual evidence. Taken together, the lower cross-label evidence overlap and stronger label-specific decision impact indicate that evidence separation and label-specific decision-making reduce evidence interference across labels, thereby contributing to the higher macro F1 achieved by the model configuration combining multi-label attention with label hypergraph reasoning.

中文翻译：为进一步评估分离后的证据是否选择性地参与对应标签决策，本文开展了标签定向删除实验（图 9）。与共享注意力对照相比，我们模型的多标签注意力模块在大多数检查场景中呈现出更强的对角线集中趋势。当删除某一标签对应的高注意力证据后，模型的决策影响更多集中于该标签自身，说明该模块能够学习具有标签特异性的视觉证据。综合较低的标签间证据重叠与更强的标签特异性决策影响，结果表明标签证据分离和标签特异性决策能够减少不同标签之间的证据干扰，从而促进采用多标签注意力与标签超图推理组合的模型配置获得更高的宏平均 F1。

To examine the contribution of label hypergraph reasoning more directly in examinations containing coexisting findings, we further stratified positive labels according to whether an examination contained a single positive target or multiple co-positive targets (Figure 10). The single-positive group was retained as a reference. In this group, the label hypergraph yielded slightly lower confidence than the ordinary label graph in some datasets, which is consistent with its design focus on higher-order relationships among coexisting diseases rather than isolated positive labels. In contrast, in the co-positive group, label hypergraph reasoning achieved the highest mean positive-label confidence across all four datasets, consistently exceeding both no label reasoning and the ordinary label graph. This pattern indicates that higher-order label interactions strengthen the integration of evidence for coexisting diseases, thereby contributing to the higher macro F1 achieved by the model configuration combining multi-label attention with label hypergraph reasoning in Table 5.

中文翻译：为更直接考察标签超图推理在共存发现检查中的作用，本文进一步按照每次检查仅包含一个阳性目标或包含多个共阳性目标，对阳性标签置信度进行分层比较（图 10）。单阳性组作为参照。在该组中，标签超图在部分数据集上的置信度略低于普通标签图，这与其主要面向共存疾病高阶关系、而非孤立阳性标签进行设

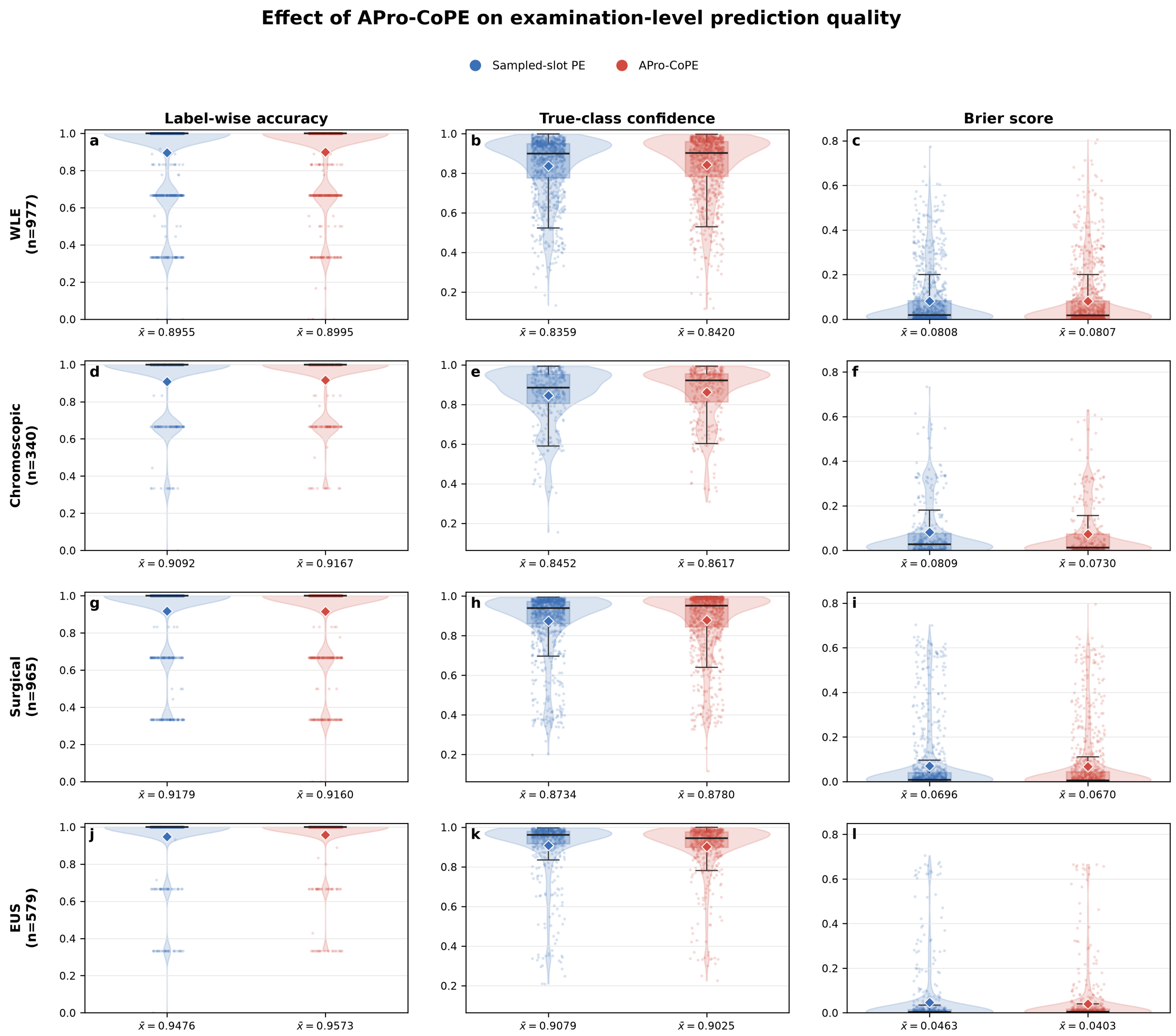

Figure 7. Effect of APro-CoPE on examination-level prediction quality relative to sampled-slot position encoding across the four gastroscopy datasets. Rows correspond to WLE, chromoscopic gastroscopy, surgical gastroscopy, and EUS; columns show label-wise accuracy, true-class confidence, and Brier score. Each point represents one examination evaluated in its held-out test fold. For examinations with more than 64 images, metrics were averaged over acquisition-ordered stratified samples covering the original sequence, with at most five sampling passes. Violin shapes show the distributions, boxes show the interquartile ranges and medians, diamonds show the arithmetic means, and $\bar{x}$ reports the corresponding mean values. Higher label-wise accuracy and true-class confidence are preferable, whereas a lower Brier score is preferable.

Table 5. Targeted ablation of multi-label attention MIL and label hypergraph reasoning across the four gastroscopy datasets using 64 sampled images per examination. Acquisition-aware visual context modeling, label-query cross-attention, and gated residual fusion were retained in all configurations. Values are macro F1 reported as mean ± standard deviation over five patient-grouped folds.

MIL aggregation	Label reasoner	WLE	Chromoscopic	Surgical	EUS
Shared attention	--	0.8840 ± 0.0148	0.8401 ± 0.0465	0.8054 ± 0.0479	0.8732 ± 0.0691
Multi-label attention	--	0.8901 ± 0.0141	0.8460 ± 0.0438	0.8110 ± 0.0447	0.8792 ± 0.0658
Multi-label attention	Label graph	0.8963 ± 0.0135	0.8518 ± 0.0406	0.8166 ± 0.0408	0.8853 ± 0.0623
Multi-label attention	Label hypergraph	**0.9149 ± 0.0114**	**0.8877 ± 0.0168**	**0.8456 ± 0.0224**	**0.9064 ± 0.0474**

MIL, multiple instance learning; Label graph denotes an ordinary learnable label graph.

计的定位一致。相比之下，在共阳性组中，标签超图推理在四个数据集上均取得最高的平均阳性标签置信度，并一致高于无标签推理和普通标签图。该结果表明，高阶标签交互能够增强模型对共存疾病证据的整合，从而促进表5中多标签注意力与标签超图推理组合获得更高的宏平均 F1。

Ablation of label-conditioned cross-modal retrieval and adaptive residual fusion To disentangle the individual and joint contributions of label-conditioned textual retrieval

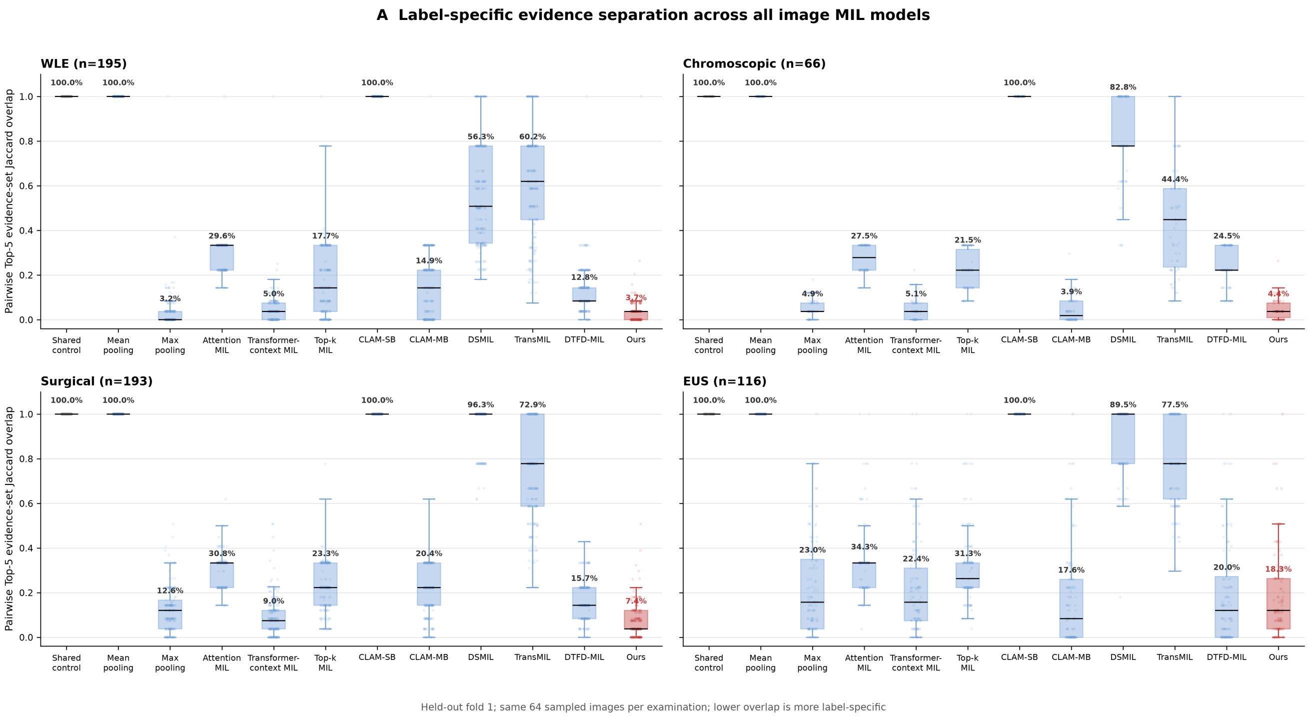

Figure 8. Label-specific image-evidence separation across the compared MIL architectures. For each held-out fold-1 examination, the same set of up to 64 uniformly sampled images was ranked separately for each of the three labels, and the mean pairwise Jaccard overlap among their Top-5 evidence sets was calculated. Each point represents one examination, boxes show the interquartile ranges and medians, and percentages report arithmetic means. Lower overlap indicates stronger separation of label-specific supporting evidence.

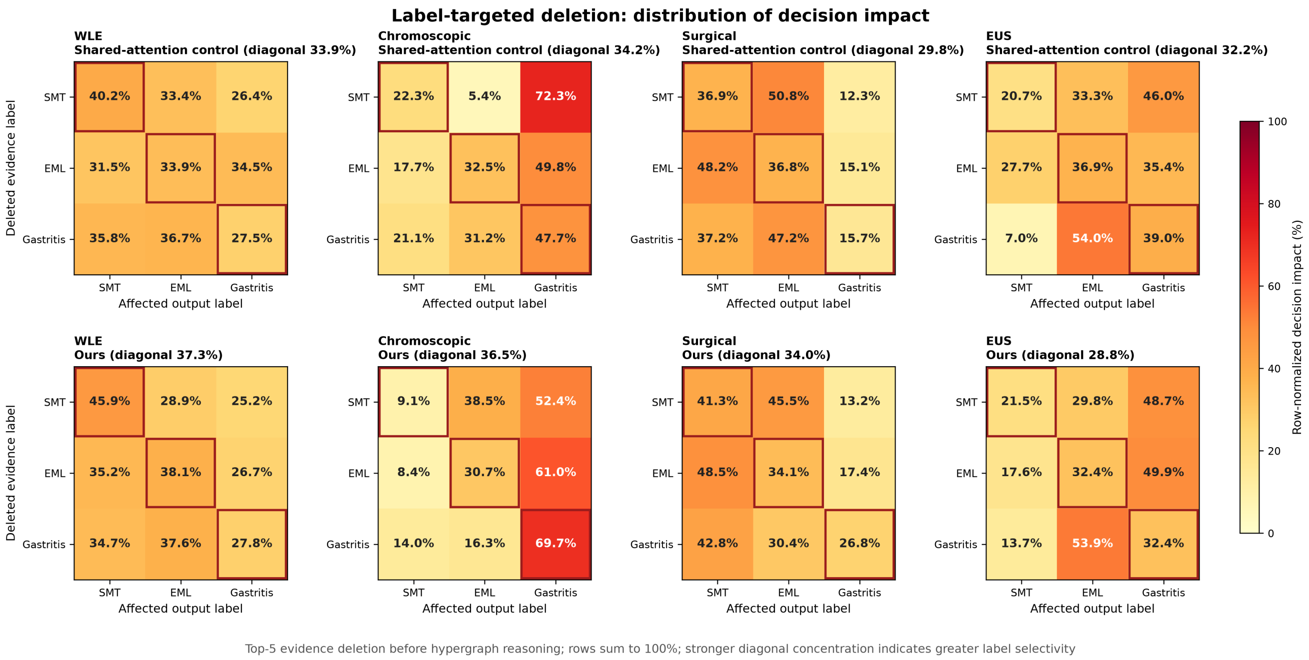

Figure 9. Label-targeted deletion analysis across the four gastroscopy datasets. For each deleted-evidence label (rows), its Top-5 attended images were removed before label hypergraph reasoning, and the resulting decision impact was distributed across the three output labels (columns) and normalized to sum to 100% within each row. The upper panels show the shared-attention control, and the lower panels show the proposed model. Red outlines mark matching deleted and affected labels; stronger diagonal concentration indicates greater label-specific decision selectivity.

and adaptive residual fusion, we conducted a targeted 2×2 ablation of label-query cross-attention (LQCA) and gated residual fusion (GRF). The experiment compared a baseline without either component with configurations enabling LQCA alone, GRF alone, or both components. The results are summarized in Table 6.

Label-query retrieval and gated fusion each improved over the visual pathway, while their combination achieved the

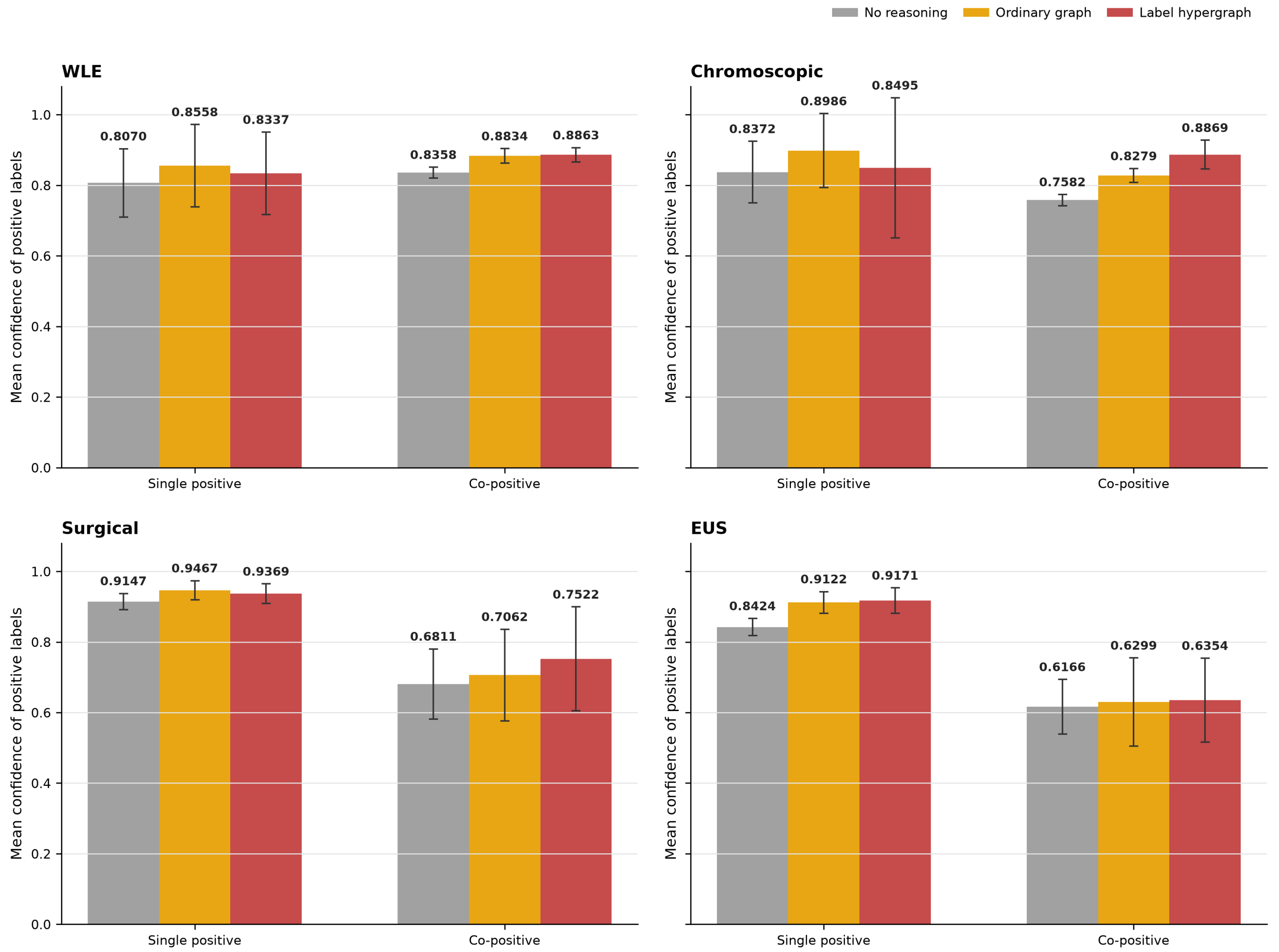

Figure 10. Positive-label confidence under different label-set complexities across the four gastroscopy datasets. Examinations were divided into single-positive and co-positive groups according to the number of positive target labels. Bars show the mean confidence assigned to positive labels, and error bars indicate 95% confidence intervals. The compared configurations used multi-label attention and differed in their label-reasoning mechanism.

highest macro F1 on every dataset, with gains of 2.42–4.40 percentage points. These results support the complementary roles of label-conditioned retrieval and adaptive control of the retrieved textual evidence.

中文翻译：为区分标签条件文本检索与自适应残差融合各自及联合产生的贡献，本文针对标签查询交叉注意力（LQCA）和门控残差融合（GRF）开展 2×2 消融实验。实验以两个组件均未启用的配置为基线，分别考察仅启用 LQCA、仅启用 GRF 以及同时启用二者的配置。实验结果汇总于表 6。

标签查询检索和门控融合均优于视觉路径，二者联合时在所有数据集上取得最高宏平均 F1，提升幅度为 2.42–4.40 个百分点。上述结果支持标签条件检索与对所检索文本证据进行自适应控制具有互补作用。

To further determine whether the improvement associated with LQCA reflected label-specific textual retrieval, we compared positive-label confidence under three retrieval conditions across the four gastroscopy datasets (Figure 11). The correct condition retained the text representation retrieved by the corresponding label query, the cross-label condition reassigned retrieved representations across labels, and the pooled condition used a shared description representation. Positive-label confidence was highest under the correct label-query correspondence and decreased after cross-label reassignment or shared pooling. This pattern indicates that disease-matched label queries retrieve more effective label-specific textual evidence, thereby supporting the macro F1 improvement obtained by the model configuration using label-query cross-attention in Table 6.

中文翻译：为进一步判断 LQCA 带来的性能提升是否源于标签特异性的文本检索，本文比较了四个胃镜数据集中三种检索条件下的阳性标签置信度（图 11）。正确条件保留对应标签查询所检索的文本表征，跨标签条件在不同标签之间互换检索表征，共享池化条件则使用统一的所见描述表征。正确标签查询对应关系下的阳性标签置信度最高，而跨标签互换或共享池化后均出现下降。该结果表明，与疾病匹配的标签查询能够检索到更有效的标签特异性文本证据，从而促进表 6中采用标签查询 cross-attention 的模型配置获得更高的宏平均 F1。

Full-factorial ablation of the main architectural components To evaluate the individual contributions and interactions of the main architectural components, we tested all 16 combinations of APro-CoPE-based acquisition-aware Transformer context modeling, label hypergraph reasoning, label-query cross-attention, and gated residual fusion using 64 sampled images per examination. When label hypergraph reasoning was disabled, an ordinary learnable label graph was

Table 6. 2×2 ablation of label-query cross-attention and gated residual fusion across the four gastroscopy datasets using 64 sampled images per examination. APro-CoPE-based acquisition-aware context modeling and label hypergraph reasoning were retained in all configurations. Values are macro F1 reported as mean $\pm$ standard deviation over five patient-grouped folds.

Text retrieval	Residual fusion	WLE	Chromoscopic	Surgical	EUS
--	--	0.8907 ± 0.0110	0.8437 ± 0.0524	0.8055 ± 0.0600	0.8769 ± 0.0675
LQCA	--	0.9031 ± 0.0128	0.8658 ± 0.0320	0.8267 ± 0.0456	0.8921 ± 0.0567
--	GRF	0.9056 ± 0.0119	0.8693 ± 0.0437	0.8299 ± 0.0491	0.8940 ± 0.0538
LQCA	GRF	**0.9149 ± 0.0114**	**0.8877 ± 0.0168**	**0.8456 ± 0.0224**	**0.9064 ± 0.0474**

LQCA, label-query cross-attention; GRF, gated residual fusion.

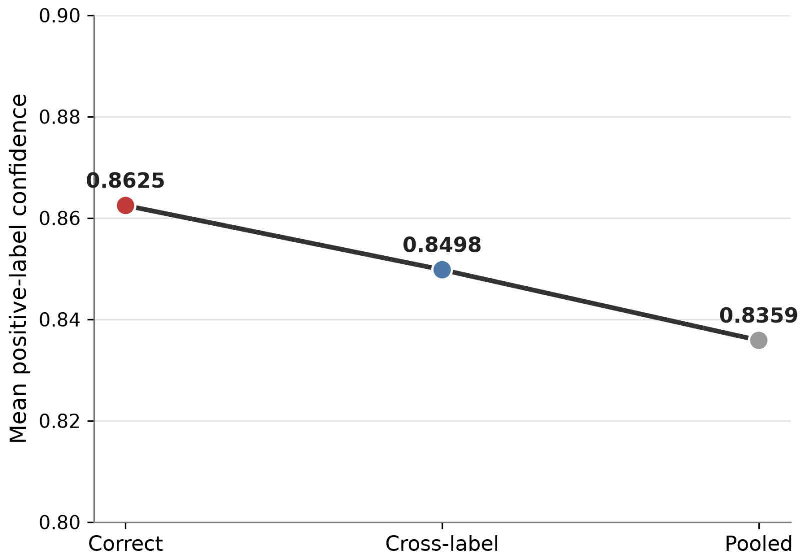

Figure 11. Label-specific textual retrieval analysis across the four gastroscopy datasets. Correct retains the text representation retrieved by its corresponding label query, Cross-label reassigns retrieved representations across labels, and Pooled replaces them with a shared description representation. The line shows mean confidence assigned to positive labels after combining the four datasets.

used. When both label-query cross-attention and gated residual fusion were disabled, prediction used the visual pathway; cross-attention without gated fusion was followed by direct addition, whereas gated fusion without cross-attention used a pooled finding-description representation.

In this experiment, M1 denotes APro-CoPE-based acquisition-aware Transformer context modeling, M2 denotes label hypergraph reasoning, M3 denotes label-query cross-attention, and M4 denotes gated residual fusion. The results in Table 7 show that APro-CoPE-based acquisition-aware Transformer context modeling alone improved macro F1 by 0.40–1.80 percentage points across the four datasets. Label hypergraph reasoning provided consistent gains of 0.70–1.37 percentage points, whereas label-query cross-attention and gated residual fusion produced larger individual gains of 1.61–2.93 and 1.76–3.36 percentage points, respectively. The combination of label hypergraph reasoning and gated residual fusion was the strongest two-component configuration on all four datasets. Enabling all four components achieved the highest macro F1 in every dataset and improved over the no-component baseline by 2.78, 5.22, 4.51, and 3.34 percentage points on WLE, chromoscopic gastroscopy, surgical gastroscopy, and EUS, respectively.

中文翻译：为评价主要架构组件各自的贡献及其交互作用，本文在每次检查采样 64 张图像的条件下，测试基于 APro-CoPE 的采集感知 Transformer 上下文建模、标签超图推理、标签查询 cross-attention 和门控残差融合四个组件的全部 16 种组合。关闭标签超图推理时使用普通可学习标签图；同时关闭标签查询 cross-attention 和门控残差融合时采用视觉路径进行预测；使用 cross-attention 但不使用门控融合时在 cross-attention 后直接相加，使用门控融合但不使用 cross-attention 时则对池化所见描述表征进行门控融合。

在本实验中，M1 表示基于 APro-CoPE 的采集感知 Transformer 上下文建模，M2 表示标签超图推理，M3 表示标签查询 cross-attention，M4 表示门控残差融合。表 7的结果表明，相对于不使用四个组件的基线，单独启用基于 APro-CoPE 的采集感知 Transformer 上下文建模在四个数据集上使宏平均 F1 提高 0.40–1.80 个百分点。标签超图推理带来 0.70–1.37 个百分点的稳定提升，而标签查询 cross-attention 和门控残差融合的单独增益更大，分别达到 1.61–2.93 和 1.76–3.36 个百分点。标签超图推理与门控残差融合的组合在四个数据集上均为表现最好的双组件配置。全部四个组件同时启用时，各数据集均取得最高宏平均 F1，并相对于无组件基线在 WLE、染色胃镜、手术胃镜和 EUS 上分别提高 2.78、5.22、4.51 和 3.34 个百分点。

Table 7. Full-factorial ablation of four main architectural components across the four gastroscopy datasets using 64 sampled images per examination. Values are macro F1 reported as mean ± standard deviation over five patient-grouped folds. M1: APro-CoPE-based acquisition-aware Transformer context modeling; M2: label hypergraph reasoning; M3: label-query cross-attention; M4: gated residual fusion. The internal position-design choices within M1 are analyzed in Table 4.

M1	M2	M3	M4	WLE	Chromoscopic	Surgical	EUS
--	--	--	--	0.8871 ± 0.0124	0.8355 ± 0.0508	0.8005 ± 0.0734	0.8730 ± 0.0715
✓	--	--	--	0.8976 ± 0.0134	0.8535 ± 0.0481	0.8140 ± 0.0639	0.8770 ± 0.0839
--	✓	--	--	0.8941 ± 0.0134	0.8492 ± 0.0583	0.8110 ± 0.0667	0.8816 ± 0.0581
--	--	✓	--	0.9032 ± 0.0115	0.8648 ± 0.0560	0.8252 ± 0.0563	0.8935 ± 0.0633
--	--	--	✓	0.9047 ± 0.0135	0.8691 ± 0.0479	0.8304 ± 0.0619	0.8953 ± 0.0711
✓	✓	--	--	0.8907 ± 0.0110	0.8437 ± 0.0524	0.8055 ± 0.0600	0.8769 ± 0.0675
✓	--	✓	--	0.8964 ± 0.0107	0.8539 ± 0.0381	0.8149 ± 0.0457	0.8851 ± 0.0668
✓	--	--	✓	0.8981 ± 0.0124	0.8532 ± 0.0448	0.8161 ± 0.0621	0.8874 ± 0.0626
--	✓	✓	--	0.9017 ± 0.0119	0.8640 ± 0.0385	0.8236 ± 0.0538	0.8919 ± 0.0677
--	✓	--	✓	0.9063 ± 0.0098	0.8693 ± 0.0444	0.8297 ± 0.0450	0.8952 ± 0.0700
--	--	✓	✓	0.8977 ± 0.0141	0.8577 ± 0.0427	0.8201 ± 0.0621	0.8884 ± 0.0587
✓	✓	✓	--	0.9031 ± 0.0128	0.8658 ± 0.0320	0.8267 ± 0.0456	0.8921 ± 0.0567
✓	✓	--	✓	0.9056 ± 0.0119	0.8693 ± 0.0437	0.8299 ± 0.0491	0.8940 ± 0.0538
✓	--	✓	✓	0.8963 ± 0.0135	0.8518 ± 0.0406	0.8166 ± 0.0408	0.8853 ± 0.0623
--	✓	✓	✓	0.8990 ± 0.0107	0.8607 ± 0.0424	0.8223 ± 0.0389	0.8881 ± 0.0527
✓	✓	✓	✓	**0.9149 ± 0.0114**	**0.8877 ± 0.0168**	**0.8456 ± 0.0224**	**0.9064 ± 0.0474**

Image-only Knowledge Transfer

To evaluate whether multimodal knowledge learned from paired images and finding descriptions could improve prediction when finding-description text was unavailable at inference, we compared four training configurations of the auxiliary image-only branch. The image-only baseline was optimized using only $\mathcal{L}_{img}$. Knowledge distillation (KD) added the prediction-level signal from the pretrained and frozen multimodal teacher, whereas multimodal branch supervision added $\mathcal{L}_{mm}$ to update the shared visual encoder. The combined configuration used both transfer mechanisms. For all four configurations, validation and test predictions were produced by the image-only branch using endoscopic images alone.

The results in Table 8 show that KD alone provided consistent but modest improvements over the image-only baseline, increasing macro F1 by 0.51–0.59 percentage points across the four datasets. Multimodal branch supervision produced larger gains of 1.58–1.82 percentage points. Combining KD with multimodal branch supervision achieved the strongest image-only results, improving over the baseline by 2.29–2.64 percentage points.

中文翻译：为评价图像与所见描述联合学习得到的多模态知识能否在推理阶段不提供所见描述文本时改善预测，本文比较辅助纯图像分支的四种训练配置。纯图像基线仅使用 $\mathcal{L}_{img}$ 进行优化；知识蒸馏（KD）进一步引入来自预训练且冻结多模态教师的预测级信号；多模态分支监督则加入 $\mathcal{L}_{mm}$ 以更新共享视觉编码器；联合配置同时使用两种迁移机制。四种配置在验证和测试阶段均只通过纯图像分支输入内镜图像进行预测。

表 8的结果表明，单独使用 KD 相对于纯图像基线在四个数据集上均取得稳定但较小的提升，宏平均 F1 提高 0.51–0.59 个百分点。单独使用图文分支监督带来更明显的提升，增幅为 1.58–1.82 个百分点。联合使用 KD 与图文分支监督取得最佳纯图像结果，相较基线提高 2.29–2.64 个百分点。

Table 8. Five-fold performance of the auxiliary image-only branch under different training-time multimodal knowledge-transfer strategies. MM sup. denotes direct supervision of the trainable multimodal branch; KD denotes knowledge distillation from a pretrained and frozen multimodal teacher. The first row uses image-only supervision without multimodal guidance. All configurations use images alone during validation and testing. Values are macro F1 reported as mean ± standard deviation over five patient-grouped folds.

Model	KD	MM sup.	WLE	Chromoscopic	Surgical	EUS
Image-only Branch	--	--	0.8841 ± 0.0068	0.7720 ± 0.0645	0.7693 ± 0.0495	0.8527 ± 0.0692
	✓	--	0.8900 ± 0.0104	0.7771 ± 0.0542	0.7744 ± 0.0653	0.8583 ± 0.0506
	--	✓	0.9023 ± 0.0069	0.7878 ± 0.0870	0.7851 ± 0.0627	0.8702 ± 0.0697
	✓	✓	**0.9105 ± 0.0120**	**0.7950 ± 0.0842**	**0.7922 ± 0.0871**	**0.8781 ± 0.0656**

Discussion

Text-only models substantially outperformed image-only MIL methods on most datasets, especially on EUS. This indicates that the gastroscopic finding descriptions contain considerably more target-related information than the image sequences under examination-level weak supervision. It also raises a potential semantic label-leakage concern: although the final labels were clinically reviewed and direct category expressions were masked, descriptions of lesion morphology and location may still strongly imply the corresponding diagnosis. In contrast, after multimodal branch supervision and knowledge distillation, the auxiliary image-only branch approached the strongest text-only performance across the four datasets and exceeded it on WLE. This comparison suggests that the ordered image sequences retain substantial discriminative information, that multimodal training enables the visual pathway to capture complementary disease-related semantics, and that the visual modality contributes meaningfully to the final prediction.

中文翻译：在大部分数据集上，纯文本模型的性能明显高于纯图像 MIL 方法，其中 EUS 上的差距最大。这说明在检查级弱监督条件下，胃镜所见描述包含的目标相关信息明显多于图像序列，同时也提示可能存在语义标签泄漏。尽管最终标签经过临床审核，且直接类别表述已被掩码，病灶形态和位置描述仍可能较强地暗示相应诊断。相比之下，经过多模态分支监督与知识蒸馏后，辅助纯图像分支在四个数据集上的性能均接近表现最好的纯文本模型，并在 WLE 上实现超越。这一对比表明，有序图像序列本身保留了充分的判别信息，多模态训练能够帮助视觉路径获取互补的疾病相关语义，视觉模态也对最终预测作出了实质性贡献。

The image-only transfer experiments showed that multimodal knowledge learned from paired images and finding descriptions improved visual prediction under image-only inference. Direct supervision of the trainable multimodal branch produced larger gains than knowledge distillation alone, while their combination achieved the best image-only performance on all four datasets. These results indicate that shared-encoder supervision and prediction-level distillation transfer complementary multimodal knowledge to the image-only branch.

中文翻译：纯图像迁移实验表明，图像与所见描述联合学习得到的多模态知识能够改善纯图像推理。可训练多模态分支的直接监督比单独使用知识蒸馏带来更大提升，二者联合在四个数据集上均取得最佳纯图像性能。该结果表明，共享编码器监督与预测级蒸馏能够向纯图像分支传递互补的多模态知识。

Examination-level multimodal multi-label classification remains an emerging clinical AI task. It requires a single model to integrate multiple images acquired during one examination, learn from examination-level supervision, identify coexisting diseases, and incorporate paired clinical text. When this study was initiated, our search of the publicly available literature found no work combining these four elements within a single task. By formulating and evaluating this setting across four gastroscopy datasets, this study establishes an initial methodological basis for examination-level multimodal coexistence classification and provides a reference for analogous tasks beyond endoscopy.

中文翻译：检查级多模态多标签分类仍是一项新兴的临床人工智能任务，需要模型同时整合一次检查采集的多张图像、利用检查级监督、识别共存疾病并融合配套临床文本。在本文研究开展阶段，据公开文献检索，尚未见在同一任务中结合上述四个要素的研究。本文在四个胃镜数据集上建立并评价这一任务设定，为检查级多模态共存分类提供了初步的方法学基础，也为内镜之外的类似任务提供了参考。

Conclusions

We developed *AMEF-MIL* for examination-level multi-label classification of gastroscopy examinations using ordered endoscopic image sequences and category-masked gastroscopic finding descriptions. Under patient-grouped five-fold evaluation, the primary multimodal output achieved mean macro F1 scores of 0.9149, 0.8877, 0.8456, and 0.9064 on the WLE, chromoscopic gastroscopy, surgical gastroscopy, and EUS datasets, respectively, and outperformed the compared image-only, text-only, and multimodal methods on all four datasets. The image-budget and position-component experiments showed that APro-CoPE improved the use of ordered examination images, while the full-factorial ablation supported the combined contribution of acquisition-aware context modeling, label hypergraph reasoning, label-query cross-attention, and gated residual fusion. For inference without finding-description text, multimodal branch supervision and frozen-teacher knowledge distillation both improved the auxiliary image-only branch, with their combination achieving the strongest image-only performance. Overall, this study establishes an examination-level multimodal multi-label classification paradigm that treats each examination as an integrated learning unit, jointly models multiple examination images, paired finding descriptions, and coexisting disease predictions, and provides a transferable framework for other multi-image clinical examinations.

中文翻译：本文提出 *AMEF-MIL*，利用有序内镜图像序列和经过类别名称掩码的胃镜所见描述进行胃镜检查级多标签分类。在患者分组五折评价中，主要多模态输

出在 WLE、染色胃镜、手术胃镜和 EUS 数据集上的平均宏平均 F1 分别达到 0.9149、0.8877、0.8456 和 0.9064，并在四个数据集上均优于所比较的纯图像、纯文本及多模态方法。图像预算和位置组件实验表明，APro-CoPE 能够改善模型对有序检查图像的利用；完整全因子消融则支持采集感知上下文建模、标签超图推理、标签查询 cross-attention 和门控残差融合的联合贡献。在推理阶段不提供所见描述文本时，多模态分支监督和冻结教师知识蒸馏均能改善辅助纯图像分支，二者联合取得最佳纯图像性能。总体而言，本文建立了以一次检查为完整学习单元的检查级多模态多标签分类范式，将多张检查图像、配套所见描述与共存疾病预测纳入统一建模，并为其他多图像临床检查提供了可迁移的研究框架。

Author contributions

Zijian Lin conceived the study, designed the computational framework, implemented the data processing and model training pipeline, performed the experiments, analyzed the results, and drafted the manuscript. Chengze Cai contributed to model design, experimental planning, and manuscript revision. Chenxi Huang provided academic supervision and methodological guidance. Hua Gao and Jie He provided clinical supervision, data access coordination, and interpretation of clinical relevance. All authors reviewed and approved the manuscript.

中文翻译：林子建构思本研究，设计计算框架，实现数据处理和模型训练流程，完成实验，分析结果并起草论文。蔡成泽参与模型设计、实验规划和论文修改。黄晨曦提供学术监督和方法学指导。高华和何杰提供临床监督、数据获取协调以及临床意义解释。所有作者均审阅并批准本文。

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Statements and declarations

Ethical considerations

This retrospective study used de-identified clinical archive data provided with permission from Zhongshan Hospital, Fudan University, and Zhongshan Hospital (Xiamen), Fudan University. The analysis did not use directly identifiable personal information.

中文翻译：本回顾性研究使用经复旦大学附属中山医院和复旦大学附属中山医院厦门医院授权提供的去标识化临床归档数据。分析过程中未使用可直接识别个人身份的信息。

Consent for publication

Not applicable because no directly identifiable personal details are reported in this draft manuscript.

中文翻译：不适用，因为本稿件未报告可直接识别个人身份的信息。

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

中文翻译：作者声明不存在与本文研究、作者身份和/或发表相关的潜在利益冲突。

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Data availability

The raw gastroscopy images and clinical report data are not publicly available because they may contain sensitive clinical information and are subject to the participating hospitals' data governance requirements. De-identified derived data tables and code may be made available from the corresponding author on reasonable request and with permission from both hospitals.

中文翻译：原始胃镜图像和临床报告数据由于可能包含敏感临床信息，并受两家参与医院的数据治理要求约束，因此不公开提供。经去标识化的衍生数据表和代码可在合理请求且获得两家医院许可后向通讯作者获取。

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